Specification, version 4

What the follow-up extracts proved, what has been decided, how E-Banker is ingested, the user interface, the audit workbooks, the macro statistics, the forward-looking adjustments and scenarios, the Mega Farm loans, and what is left to build

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Working document, updated 13 October 2026 (section 3.10, the opening of the revenue roll-forward, facilities drawn in tranches, floating-rate re-estimation and arrears at the opening; decisions D32 to D39; the forward-looking module now specified in its own document). Earlier update 11 October 2026 (section 3.7, the fee ledgers and the true-up into 4206). Replaces version 3 of 24 September 2026 as the single point of reference; version 3 is kept in 9. Archive and still applies wherever this document is silent. Where the two differ, this one governs.

0. How to read this document

Version 3 was written before MAIIC's database could be read directly. It described the engine around three extracts (A, B and C) that MAIIC's ICT team produced with their own scripts, and it listed twenty-three choices that were still open. Between 5 and 7 October 2026 Barry Makumba ran twenty-seven data-dictionary queries and eighteen follow-up extracts for us, then five more on request. Those extracts are the whole loan ledger and every supporting table, and they changed the picture: the data gaps of version 3 are closed, the arithmetic of the core system is reproduced to the cent, and most of the open choices can now be settled on evidence rather than on argument.

This document records that. It is written, as before, for three readers at once: the CFO, who has to be confident that the numbers will stand up to Deloitte; the ICT manager, who supplies the data and hosts the system; and the developers, who build it. Every section starts in plain language. The same three words are used in the same way as in version 3:

- **Agreed** means a decision that has been taken, by whom and when.
- **Decided and seeded** is new. It means a choice that has been settled by Dupleix on the evidence, written into the Governance Centre as the default in force, and is waiting only for Dr Thom to confirm or change it. Section 4 lists every one, with its default.
- **Proven** means a fact tested against MAIIC's own data and reproduced, with the file and the count given so that anyone can repeat the test.

Where a file is named, it is one of the extracts committed under `docs/bootstrap/` in the repository (section 6.10), with a copy in `2. Documents from clients\Raw Query Scripts\Query Requests to MAIIC\Follow-Up Scripts Resutls\`. The scripts that produced every figure in this document are in `Build files\` beside them, with a README that says which script makes which number.

Where a section says a design is taken from the Dupleix suite (sections 6, 11, 12, 13, 14 and 15), the code it was taken from is copied verbatim under `docs/reference/` in the repository, indexed by section in its README and pinned to the source commits in its manifest, so that the completeness of this document can be verified against working code and the build can port rather than reinvent. That folder is code only. The regulatory texts the document relies on are under `docs/regulatory/` (section 3.6).

1. What has happened since version 3

When	What
25 Sep 2026	Progress meeting. Phases P1 to P4 built and merged: the Governance Centre, reference rates and spreads, contractual interest and the reconciliation, schedules.
5 Oct 2026	MAIIC's three extract scripts analysed. The data-dictionary pack of 27 queries written and run by Barry the next morning. Codex reviewed the analysis; eight corrections adopted.
6 Oct 2026	18 follow-up extracts specified, with a run order and session settings that make every date unambiguous. Barry ran all 18 by 10 am on 7 October.
8 Oct 2026	Pack 2 specified (the September month-end feed, the interest arithmetic, collateral, guarantors, the origination payloads, GL_03, the Mega Farm schemes); seven of its eleven runs received the same evening and checked: the first monthly feed ties on every account, GL_03 settles the year-end adjustments, the collateral register is in hand.
12 Oct 2026	Section 3.7 clarified (MAIIC): the 259,317,000 was a manual adjustment recommended by the auditors, who extrapolated from a sample; it was applied in the TB-to-AFS mapping and not posted in E-Banker or the trial balance (Dr Thom's workbook, typed values in both sheets).
12 Oct 2026	`TB_03` (pack 4 RUN 9c): the trial balance rows from E-Banker for December 2025, August and September 2026; if the first two tie to Finance's, September's trial balance comes from the database (sections 3.9 and 7.7).
12 Oct 2026	Section 3.9 added: reschedules rebuilt from E-Banker's own tables (superseded instalment rows, four dated copies of the chart, `MODIFICATION_LOG`) through `RS_02a` to `RS_02g`, pack 4 RUN 8; the stored loan books ruled out as a register (4 weak candidates); probes of `TRIAL_BALANCE_HISTORY` and `CUMOVUCH_OLD` (RUN 9).
12 Oct 2026	Section 3.8 added: status H is written off and closed, on the evidence of the balances, the 29 June 2026 transfer to GL 1064 and Finance's loan books; the write-offs of 2024 and 2026 traced in GL 1064; the rule that a write-off is never read as a collection; `GL_05` requested. Section 2.1 corrected (H was read as paid-up awaiting closure).
11 Oct 2026	Section 7.8 added: how fees are ingested and reconciled, from E-Banker to the EIR, with the sources and their precedence, the five delivery routes of `GL_04`, the link of each fee to its loan by the batch, the checks, the settings and Finance's steps.
11 Oct 2026	Built: the fee GL check on every fee import and on Fee Classification, by month and year, with a Governance Centre switch (`fee_gl_reconciliation`) and a fee-only import permission (`eir.fees.import`) for Finance (section 7.4).
11 Oct 2026	The fee income ledgers (4871, 4873) reconciled to the trial balance in every month of 2025 and 2026 to date, and the 2025 year-end true-up into 4206 traced from E-Banker to the audited accounts: 21,865,867.35 of suspense reversals and 259,317,000 of arrangement fees presented as EIR interest (section 3.7).
7 Oct 2026	The 18 extracts audited in full. Five further queries run the same day to explain two reconciling differences. The take-on amortisation schedules mapped to E-Banker accounts. The open choices of version 3 written into the Governance Centre with seeded defaults. Two of them decided: which record is contractual (O19) and how a mid-month PLR change is charged (O2).
13 Oct 2026	The forward-looking module rebuilt to its own specification (`MAIIC_FLI_Module_Specification_v1_2026-10-13.md`): model registry, month record per scenario, sign-off. Two UI audits closed (imports to reports, create, edit, delete and refresh on every screen; Maintenance reduced to seven jobs). The revenue roll-forward corrected for its first month, for facilities drawn in tranches, for floating-rate loans whose rate has moved and for arrears at the opening (section 3.10); amortised cost now ties to the ledger within 5.8 million on 11.47 billion at August 2026 and the book's EIR revenue shift for January to August 2026 is -117.7 million.

The result in one sentence: **the engine's inputs are now complete, every reconciliation either ties to the cent or has a named cause and a named fix, and what remains is engineering plus a short list of answers that only people at MAIIC can give.**

2. The data the engine now runs on

2.1 What replaced Extracts A, B and C

Version 3 built the engine around a contract master (Extract A), a transaction file (Extract B) and an interest file (Extract C), each produced by a MAIIC script with its own faults: a fixed/floating label that was the wrong way round, amounts with their signs stripped, and a 2025 that was counted three times. None of the three is needed any more. The engine reads the tables behind them.

Need	Version 3 source	Now	What is proven about it
Who the borrower is, the facility, its terms	Extract A	`P1_02` account master (184 accounts) and `P1_03` loan master, every column, with the Interest Policy and the Floating Flag read directly	184 accounts in the 18 loan schemes: 89 active (A), 46 closed (C), 25 dormant (D), 21 code H, 3 frozen (F). 109 were taken on at migration, 75 opened since.
Every posting on every account	Extract B	`P1_01` the ledger (CUMVOUCH), signed amounts, every transaction type	Ties to the month-end balance history on 2,533 of 2,549 account-months; to the stored loan book's carrying amount on 2,264 of 2,264; to the trial balance on 1050101, 1050102 and 1050401 in every month. The 16 open balance-history rows are one account (section 3.5).
Interest charged each month	Extract C	The ledger's own interest postings (type 303, plus the year-end type 120)	Equal to Extract C run 41 on all 1,564 account-months in its window and in total, MWK 5,293,988,207.06, January 2025 to July 2026. Extract C is retired.
The rate on each account over time	Not available	`P1_04` the rate set-up per account (1,291 dated rows, 183 accounts) and the monthly loan book's rate column	The charged rate is the loan book's rate, not the set-up table's (section 3.2). The set-up table dates the changes.
The prime lending rate	A hand-typed file	`P2_06` the PLR master, 49 rows, March 2020 to October 2025, with the Reserve Bank effective dates	Replaces the repaired file of 24 September.
The monthly loan book	Excel reports, Dec 2025 to Aug 2026 only	`P2_08` the stored loan book, every month-end from December 2024 to August 2026	21 month-ends; 2,361 unique account-months after keeping the latest run of each month (section 6). Closes open choice O8.
Month-end balances	Not available	`P2_09` the balance history	Independent of the ledger and agrees with it, which is how the ledger was proven.
Fees	Not available (the largest gap in version 3)	`P2_07` the disbursement charges table: 3,651 rows, 52 in scope, with the income GL of each fee; and, once re-run, `P2_13` the origination system's sanction and disbursement payloads	Covers loans disbursed through E-Banker. Fees on the 109 take-on loans are not in any table and come from Tamanda's template (agreed O14). The first `P2_13` export carried the payload as the literal text CLOB; the re-run renders it as text.
Collateral and guarantors	Not available	`P2_11` the customer security register (118 distinct securities, MWK 47.1 billion, on 86 accounts) and `P2_12` the guarantors (71 accounts)	The security register repeats a security across a customer's accounts, so it is de-duplicated on its own id before any coverage is computed. 58 of the 89 active accounts have a security on file; the 31 that do not are a list for Credit before the loss-given-default is built on it.
Drawdowns	Not available	`P3_12` the disbursement schedules	58 of 314 facility rows have between 2 and 10 tranches.

Need	Version 3 source	Now	What is proven about it
The repayment pattern	The instalment chart	P2_10 the current chart and P3_11 the instalment plans	The chart exists for 75 of 184 accounts, and actual repayments are within 5 percent of it on 23 of 69 testable accounts. The engine generates the expected pattern from the loan's terms (section 7.5) and keeps the chart as a reference.
Status changes	Not available	P3_13 every status change with its reason	Explains codes D and F. Code H (21 accounts) marks accounts written off and closed: "Paid Up" in the history, balances unchanged until a transfer to GL 1064 Term Loans Write Offs on 29 June 2026 (section 3.8; Finance to confirm).
Daily accrual	Not available	P3_14 the daily interest accrual, 7,489 rows from 1 November 2024	Held for 14 accounts only. For those it proves the month-end posting is the sum of the daily rows, to the cent.
GL control	The trial balance	The trial balance, every month, plus GL_02 the keyed GL opening balances and GL_03 the GL legs of the year-end batches; for fees, the GL spools of 4871 and 4873 (Tamanda, 19 August) and from pack 4 GL_04 the fee GL legs straight from E-Banker	Interest income on a year-to-date basis ties on 83 of 83 comparable account-months, and December 2025 to the cent once the year-end adjustments are read correctly (section 3.4). The fee GLs tie to the trial balance in every month of 2025 and 2026 to date (section 3.7).
The monthly feed	Not tested	M09_01, M09_02 of 8 October: the ledger since the last loaded id with August re-pulled, and the month-end balances for August and September 2026	The first run of the feed contract of section 6.4: August identical to the 7 October file with no back-dated posting; September's 118 postings on 83 accounts; the balance change equals the ledger on 105 of 105 accounts.

2.2 Two tables that do not help, and why

INTEREST_SUMMARY (P1_05) looked like the calculation behind every interest posting. It is a working table that the month-end run rebuilds; it holds the latest month only (82 rows, all August 2026), confirmed by a second run on 7 October. INTEREST_PRODUCT (P1_05b, 8 October) does keep history, October 2024 to September 2026 on 126 accounts, and its monthly totals equal the ledger's interest postings on 1,763 of 1,818 shared account-months; but its columns are not what their names say: the "product" column holds the computed interest, not the balance times days, and the days column is zero. It confirms the postings and gives a month's interest before the ledger is pulled; the arithmetic behind each posting still comes from the posting's own narration, which states the period and the rate (section 3.1), and from the daily accrual where it exists.

The instalment chart (P2_10) is current, not historical, and marks a missed instalment as paid. It is a reference, not a cash-flow source.

2.3 The take-on loans

The 109 loans that existed before E-Banker went live were taken on at 31 July 2024, 77 of them with a single posting on that day totalling MWK 8,297,388,309.25. Their history before that date is not in E-Banker. Tamanda's amortisation schedules at 31 October 2024 (100 blocks) have been mapped to the E-Banker accounts: 105 of 109 facilities matched, 104 of them with high confidence, the principal equal to the take-on posting on 77 and the value date equal to the first disbursement-schedule date on 82. The four not matched are Favoured Farms 2022 and three equity-type positions that the engine must refuse in any case (MMC, Wi Jays, Good Hope). The workbook is with Tamanda to tick; its Upload summary sheet is the take-on loader's input (section 7.6).

3. What is proven about E-Banker's arithmetic

Everything in this section was tested on the ledger. The engine's contractual-interest service reproduces it, and the reconciliation names a cause for any month that does not agree.

3.1 Interest is charged for the whole calendar month at one rate

Every month-end interest posting carries a narration of the form "To Trf IntDR Fm 01-JUN-25 To 30-JUN-25 @ 32.65%". Reading all 2,047 of them:

- The period runs from the 1st to the month-end in every ordinary month (299 of the 303 months that had a rate change inside them; the other four are drawdown months). The period is never split at the date the rate changed.
- Where the daily accrual exists, its daily amount is constant through the month and the month-end posting equals the sum of the daily rows to the cent.
- The amount is the previous month-end balance, including interest already capitalised, times the rate, times the days in the month, over 365. Rebuilt this way, 1,595 of 2,047 postings agree within 0.1 percent and 1,828 within 1 percent; the postings that agree within 1 percent are 91.6 percent of the value. The remainder are drawdown months, moratorium months and the 28 year-end adjustments of section 3.4.
- **When the rate changes inside a month, the whole month is charged at the new rate.** In the 229 account-months where the loan-book rate changed from the previous month-end, the narration carries the new rate in 217, the old rate in 2 and neither in 10. A PLR cut on the 20th reduces the whole month's interest. This decides open choice O2 (section 4.2).

3.2 The charged rate is the loan book's rate, not the set-up table's

The rate set-up table (`P1_04`) holds, for each account, a row for every PLR change with a `PLR_RATE`, a variation and an `INTEREST_RATE`. It is not the rate E-Banker charges. Account 000104420000005 is charged 32.45 percent through April 2025 and 32.65 percent from May while the set-up table says 30.35 falling to 30.15; the difference is constant for months at a time. The charged rate is the one on the loan master, which the monthly loan book reports: the narration equals the loan book's month-end rate on 1,453 of 1,600 testable account-months, and in every month where the two disagree the loan book was re-run later with a back-dated change.

On the accounts that behave as the product intends, the charged rate is exactly the PLR plus a fixed spread, applied in the month of the PLR change: account 000104420000010 is 25.2 + 6.0, then 24.7 + 6.0, 22.4 + 6.0, 20.8 + 6.0, 20.6 + 6.0 and 20.4 + 6.0 as the PLR fell through 2026. On others the spread moves (account 000104420000005: 7.05, 7.45, 7.55, 7.45). A moving spread is MAIIC re-pricing at its own option, which is the reset-or-modification question of open choice O17, and the engine reports every such move.

Consequence for the build: the rate-history importer takes the **monthly loan-book rate** as the rate charged, the **PLR master** as the reference rate, and the set-up table as the register of when a change was keyed. The spread is derived as charged rate less PLR, month by month, and an account whose spread drifts is flagged (the `SpreadDerivationService` of P2 already does this).

3.3 Signs, transaction types and the six GL codes

Debits are negative in the ledger. The transaction types that matter: 301 drawdown, 303 monthly interest, 305 repayment, 306 reversal of a repayment, 120 the year-end "Diff Int Credit by ROI" adjustment, 300 write-off, 343 reversal of a credit, 900 and 901 a fee posted and reversed, 400 and 401 the Nascomex equity entries. The loan accounts sit on six GL codes: 1050101 and 1050102 (MAIIC Agricultural and Industrial), 1050201 and 1050202 (FInES Agricultural and Industrial, the 10 percent concessional book), 1050401 (term loans) and 1050103.

The income GLs, and the second chart of accounts. Interest income sits on one E-Banker GL per loan GL: 4215 (1050101), 4216 (1050102), 42019 (1050401; printed as 4201 in the code column of the trial balance export, with the full code in its title), 4218 (1050201), 4219 (1050202), and 4221 to 4224, 4260 and 4262 for the Mega Farm schemes. Fees sit on 4871 legal, 4873 arrangement, 4874 insurance and 4875 PCG guarantee. Fees are not always posted to the loan account: in 2025 about 44 percent of fee income, and nearly all of it from March 2026, was credited to the fee GL in the drawdown batch with the loan account carrying the gross drawdown (section 3.7). The annual accounts are prepared in a second ledger (QuickBooks) with its own codes, and E-Banker's interest GLs roll up into it: 4215, 4216 and 42019 into 4206 Interest on Term Loans-MAIIC; 4218 and 4219 into 4209 Term loan interest-FInES (section 7.7). Any extract, reconciliation or interface to Finance has to carry that mapping.

3.4 The 28 year-end adjustments of 31 December 2025

On 31 December 2025 E-Banker re-rated the year's interest on the PLR-linked book to the rate on each account at year end and posted the difference: 22 debits (type 303) totalling MWK 96,390,096.16 charged to customers and 6 credits (type 120) totalling MWK 85,166,683.31 in their favour, on 1050101 and 1050102 only, in the same interest-posting batches as December's ordinary interest (INTP 41 and 42). The 6 credits are exactly the difference between Extracts B and C that version 3 could not explain.

`GL_03` of 8 October shows the other side: **one net voucher per batch to interest income**. Batch 41 credits 4215 by 32,956,675.55 (the 1050101 debits of 35,940,279.17 less credits of 2,983,603.62); batch 42 debits 4216 by 21,733,262.70 (the 1050102 credits of 82,183,079.69 less debits of 60,449,816.99). All 28 adjustments therefore went through 2025 interest income, net 11,223,412.85, and with that reading the December movement on both income GLs equals the ledger to the cent (1050101: ordinary interest 27,129,967.64 plus 32,956,675.55 equals the trial balance's 60,086,643.19; 1050102: 173,635,459.72 less 21,733,262.70 equals 151,902,197.02). An earlier reading that the debits had not reached income was a double count, since the debits are type 303 and sat inside the ordinary figure; it is withdrawn.

The engine treats the 28 as interest of December 2025 and shows them as their own line in the reconciliation. The remaining question for Finance is one of intent, not of posting: whether the year-end re-rating to the closing rate was meant to be the policy, since the engine's contractual interest follows the month-end rate month by month (section 3.1) and the two differ by exactly these amounts.

3.5 Two reconciling differences, both explained

The FInES GLs. The trial balance for 1050201 is MWK 400,000.00 above the sum of its accounts in every month and 1050202 is MWK 1,000,000.00 below, while the other GLs agree to the cent. Barry's queries of 7 October showed there is no posting without an account (`GL_01` is empty) and that the yearly GL opening balances are keyed by hand (`GL_02`): the 2025 opening for the two FInES GLs was keyed 400,000 above and 1,000,000 below the accounts, and the 2026 opening was taken from the trial balance, carrying the difference forward. Fix: Finance confirms which figure agreed to the 2024 audited accounts and adjusts the two GL openings. Until then the engine carries the 600,000 net as one named line.

Zaithwa Farms (000104450000015). The month-end balance table shows the customer MWK 1,827,633.62 better off than the ledger from May 2025. The ledger is complete and right (24 postings, closing at 1,225,683.03 in the customer's favour) and so is the stored loan book. The balance table holds 30 April 2025 twice, and its May row restarted from nil and omitted both May receipts; every later row inherits the error. It is the only duplicate month-end row in the whole table. Fix: the vendor rebuilds the account's rows from May 2025; Finance checks any statement sent to the customer. No change to the engine, which reads the ledger.

3.6 The Reserve Bank's directive for development finance institutions, and staging

MAIIC is licensed as a development finance institution, and the Reserve Bank of Malawi classifies its lending under the **Financial Services (Credit Risk Management for Development Finance Institutions) Directive, 2018** (Malawi Gazette Supplement of 13 July 2018, No. 18A, pages 42 to 47; a copy is under `docs/regulatory/` in the repository and beside this document). It is the authority behind MAIIC's staging, and the engine's governed thresholds follow it.

The directive sorts facilities by repayment period (section 2): short term, not more than 12 months; medium term, 12 to 60 months; long term, over 60 months. It then classifies each by how long instalments are overdue (section 10) and sets the provision for each class (the Schedule):

Class	Short-term facility	Medium- and long-term facility	Provision
Standard	current, to 30 days overdue	31 to 90 days	0 percent
Special mention	31 to 90 days	91 to 180 days	5 percent
Substandard	91 to 180 days	181 to 365 days	20 percent
Doubtful	181 to 365 days	366 to 746 days	50 percent
Loss	over 365 days	over 746 days	100 percent

Substandard, doubtful and loss are **non-performing**: the facility goes on non-accrual (section 13), interest accrued but not collected is reversed and interest is recognised only when received in cash, regardless of collateral; a loss facility is written off in the following quarter (section 12); where the directive's provision exceeds the IFRS one, the excess is an appropriation to a loan-loss reserve, not capital (section 15).

What the engine takes from it. Stage 3 (credit-impaired) is the directive's non-performing line: **181 days past due for a medium- or long-term facility, 91 days for a short-term one.** That is where MAIIC's long-standing rule of "over 181 days" comes from, and it rebuts the IFRS 9 presumption that default occurs no later than 90 days past due (B5.5.37) on the authority of the regulator's own classification of development-finance lending; the 2025 financial statements rest on it. Stage 2 is kept at 31 days past due for every tenor: the IFRS 9 presumption of a significant increase in credit risk at 30 days (B5.5.11) is not rebutted, because the directive's wider "standard" band for medium- and long-term facilities is a prudential classification, not evidence that credit risk has not risen. The rebuttal of that 30-day presumption stays available as a governed proposal, inactive until Dr Thom signs it (the staging rebuttal of O13).

The thresholds are seeded by tenor class into `staging_thresholds`, each row carrying the directive section it rests on: short term 31 and 91; medium and long term 31 and 181; the Mega Farm programme, which is seasonal input finance repaid after harvest and therefore short term whatever tenor the account carries, 31 and 91; and the long-term Stage 2 proposal at 91, future-dated. The classifier reads days past due from the stored report's overdue date and arrears buckets (section 6.2).

The three rules MAIIC lives with, compared. The signed 2025 accounting policy, the ECL model that produced the 2025 figures and the directive do not say the same thing, and the engine has to carry all three in a way the auditors can follow.

	The accounting policy (2025 statements, notes 3.8 and 22.8.1)	The November 2025 ECL model	The directive, as seeded
Stage 3, default	"when the borrower has missed four consecutive contractual payments, or the loan becomes 90 days past due", with note 3.8's door: "unless the Entity has reasonable and supportable information to demonstrate that a more lagging default criterion is more appropriate"	Over 181 days past due. Nothing at 90	Non-performing from 181 days (medium and long term), 91 (short term)
Stage 2, significant increase in credit risk	Changes in the probability of default against inception, qualitative events, the 30-day backstop	31 to 180 days past due. The probability-change test exists in the sheet but fired on none of the 100 loans	31 days (IFRS 9 B5.5.11, not rebutted)
The directive's classes	not mentioned	a column, blank on every loan	Standard, special mention, substandard, doubtful, loss, by tenor
A tenor split	none	none: 181 for every facility	91 days at 12 months or less, 181 above
November 2025		100 loans: 72, 10 and 18 by stage; exposure 11.54, 1.22 and 1.08 billion	the E-Banker book of 125 loans on the demo: 79, 25 and 21; 7.39, 4.38 and 2.06 billion

Five things follow. **The policy note says 90 days and the model applied 181**; the note's door was walked through but the note does not say so, and for the 2026 statements it must, with the directive as the reason, or the model must move to 90. **The four-consecutive-payments test** is in the policy and in neither the model nor the directive; on monthly instalments it falls at about 120 days, and the engine will carry it as a second Stage 3 trigger from the instalment plan and the ledger. **Stage 2 was, in practice, days past due alone**; the engine's 31 days reproduces that, and the probability-change test is the work of section 14. **No tenor split existed**, so the eight short-term facilities of November were staged at 181 where the directive says 91: small on the MAIIC book, decisive on the Mega Farm book. **The model held 100 loans where E-Banker held 125**; the 25 it leaves out (closed, code H, nil balances) are to be named by Finance before the two stagings can be reconciled.

Which days past due. The directive counts days from the overdue instalment. The stored report carries both the date of the oldest unpaid instalment and the ageing of the overdue amounts by bucket, and on some accounts they disagree: an account partly cured keeps an old overdue date while its buckets empty. The engine takes the instalment date, as the directive does, and shows the bucket ageing beside it; the choice is governed (`dpd_basis`) so that it can be changed with a reason.

Decision D31. The governed thresholds stand as seeded: Stage 3 at 181 days for medium- and long-term facilities and 91 for short-term and Mega Farm, Stage 2 at 31 days. They follow the directive and reproduce the 2025 model, so the 2025 figures do not move. The rebuttal of the 90-day presumption is written down as the O13 sign-off, with the directive cited by gazette and section, and a sentence is drafted for the 2026 policy note. The four-payment trigger and the days-past-due basis become governed items beside the thresholds.

Two differences between the directive and IFRS 9 that the engine must show rather than resolve: the directive stops interest on a non-performing facility, where IFRS 9 5.4.1(b) continues it on the net carrying amount (the trial balance's interest-suspense account, 1320, is the directive's rule in practice, and the reconciliation carries the difference as its own line); and the directive's provision percentages are prudential floors, so the loan-loss reserve of section 15 is where any excess over the ECL sits.

3.7 Fee income and the year-end true-up into 4206 (added 11 October 2026)

In plain language. The fees MAIIC charges at drawdown are booked in full to two income accounts, 4871 Legal Fees and 4873 Arrangement Fees. When the annual accounts are prepared, Finance moves the part of those fees that belongs in the effective interest rate, and a few interest adjustments, into one interest account, 4206 Interest on Term Loans-MAIIC. Dr Thom described this at the meeting of 24 September 2026. The figures below show exactly what moved in 2025 and where each figure was found.

The fee ledgers agree to the trial balance. Tamanda's GL spools of 19 August (`GL 4871` and `GL 4873`, as at 31 December 2025 and 31 July 2026) agree to the trial balance year-to-date in every month from January to November 2025 and from January to July 2026, to the cent. Before the closing voucher of 31 December 2025 they stand at **171,128,135.00** (legal) and **311,342,792.00** (arrangement), which are also the figures in the final E-Banker trial balance for December 2025 and in the "Mapping TBs" sheet of Dr Thom's workbook of 24 September (`Mapping TBs August 2026 to Audited Financial Statements4.xlsx`). Within the year the spools hold two corrections only: a legal fee over-deduction reversed on 24 October 2025 (net 100,000) and 7,629,800 moved from 4873 to 4871 for Prominade Medical Centre on 30 December 2025.

Not every fee touches the loan account. Each spool line was matched to the account ledger (`P1_01`) by date and amount. In 2025 about 56 percent of each fee account (legal 96,579,275 on 20 lines; arrangement about 175 million on 21 lines) has a matching fee debit on the loan account at drawdown. The rest was credited to income in the drawdown batch with no fee leg on the account: the loan account carries the gross drawdown instead (Malawi Police SACCO, 7 May 2025: 1,046,210,000 drawn, being 1,000,000,000 plus 30,000,000 arrangement and 16,210,000 legal). From March 2026 almost no fee reaches a loan account. The 23 "BRFF" fee lines of June and July 2026 have no drawdown in the eighteen schemes at all; whether BRFF is a programme outside them is asked in section 5. **So the fee importer reconciles to the fee GL spools, not to the account ledger** (section 7.4).

What 4206 carries in the 2025 accounts.

4206 Interest on Term Loans-MAIIC, 2025	MWK	Source
E-Banker 4215 Interest on MAIIC Agricultural Loans (includes the year-end adjustment +32,956,675.55, section 3.4)	554,447,006.83	GL 4215 spool; final E-Banker TB
E-Banker 4216 Interest on MAIIC Industrial Loans (includes the year-end adjustment -21,733,262.70)	1,513,020,260.19	GL 4216 spool; final E-Banker TB
E-Banker 42019 Interest on MAIIC Term Loan	39,093,897.99	Final E-Banker TB
4206 in the initial TB submitted to the auditor	**2,106,561,165.01**	AFS mapping workbook
Audit adjustment "Interest in suspense reversals" (the 1320 Interest Suspense balance)	21,865,867.35	AFS mapping workbook, Final AFS TB of 19 March 2026
4206 in the final AFS TB	**2,128,427,032.36**	Same
Arrangement fees presented as EIR interest income	259,317,000.00	Dr Thom's workbook, "Income Statement" sheet against "Mapping TBs"
EIR Interest Income line for 4206 in the audited accounts	**2,387,744,032.36**	Same

E-Banker 4218 and 4219 together (447,829,531.72) are 4209 Term loan interest-FInES, unchanged.

The fee move is a manual adjustment, not a posted journal. The final AFS TB lists every audit adjustment (31 of them) and 4873 is still 311,342,792 in it. The 259,317,000 exists only in the income-statement mapping, where 4873 shows 52,025,792 and 4206 shows the 259,317,000 more; profit is the same in both sheets (4,850,745,713). No spool will ever show it, and it is not a set of whole fee postings: no combination of the 35 lines of GL 4873 for 2025 leaves 52,025,792. It was a manual adjustment recommended by the auditors, who

tested a sample of fees and extrapolated the result (MAIIC, 12 October 2026), applied in the TB-to-AFS mapping and not posted in E-Banker or in the trial balance: in Dr Thom's workbook 4873 is 311,342,792 in the trial balance sheet and 52,025,792 in the income statement sheet, both typed values with no adjusting entry. E-Banker has no effective-interest calculation, which is why this engine is being built. Its detailed working is therefore not needed; the figure is kept only as the 2025 comparator for the engine's output. Legal fees were not moved in 2025. In 2024 the move went the other way: 37,610,000 out of 4206 into 4873, so 2024 arrangement fees are 1,299,964,206 in the ledger and 1,337,574,206 in the accounts.

What this corrects. Version 3 (fact F18 and open item O23) and the note of 24 September read 52,025,792 as the 2025 ledger figure for arrangement fees. It is the figure after the move; the ledger figure is 311,342,792. O23, which asked why 2024 arrangement fees were so much larger than 2025, is smaller than it looked: 1,299,964,206 against 311,342,792, not against 52,025,792.

What the engine does with it. The engine's inputs are the fees as charged, which are the ledger figures above. Its output for 2025 is compared with what the accounts recognised: 259,317,000 of arrangement fees treated as interest, plus the 21,865,867.35 suspense reversal, both inside 4206. The journal the engine proposes (O10, O21) replaces this presentation entry. Until Finance opens a dedicated EIR adjustment account, 4206 is the account the true-up lands in, because that is where it lands today.

Scripts: `docs/build-files/fee_gl_recon_2026-10-11.py` (the spools against the trial balances and the account ledger). The 4206 bridge is read from the AFS mapping workbook ("TB December 2025", "Final E-Banker TB Dec 2025" and "FINAL AFS TB as of 19th Mar 26") and Dr Thom's workbook. Dr Thom's email of 24 September 2026 is saved beside the workbook in `New Doc Received 21 Sep`.

3.8 Status H, and the write-offs of 2026 (added 12 October 2026)

In plain language. Twenty-one loan accounts carry the status code H in E-Banker's account master. The status history says they were paid up; the balances say they were not. In June 2026 Finance cleared what they still owed by a transfer into the write-off account, which E-Banker recorded as a repayment. Read at face value, the ledger turns MWK 38.5 million of losses into cash collected. This section sets out the evidence, what H means on it, and the rule the engine follows.

What the records show.

Source	What it shows
Account master (`P1_02`)	21 accounts with STATUS_CODE H, status reason "Paid Up", status dates 11 to 16 December 2025 (17 accounts) and 30 June 2026 (4)
Status history (`P3_13`)	No H row at all. Each of the 21 went from A to D "Paid Up" on those dates; the H is set in the master without a history entry
Balance history (`P2_09`, `M09_02`)	The balances did not change between December 2025 and May 2026: Lenziemill Milling owed 5,667,578.18, COMSIV 8,318,966.49, Mwachisosa Women Producers 6,676,759.40, and so on. "Paid Up" was not a repayment
Ledger (`P1_01`)	On 29 June 2026 one batch, TRF 3, credited MWK 38,492,032.54 to 20 of the accounts, type 305 (the repayment type), narrated "ACCOUNT CLOSURE"; no leg of the batch is on another loan account
Trial balance (Finance)	GL 1064 Term Loans Write Offs rose by exactly 38,492,032.54 between May and June 2026
E-Banker manual (section 7.4)	A genuine account closure is posted with batch type ACCLO; this batch was a TRF transfer
Finance's loan books	All 21 accounts carry balances in the May 2026 loan book and are nil from June 2026

What H means. On this evidence H marks an account **written off and closed**. Finance has been asked to confirm (section 5). The other codes, for the record: A active; C closed (15 of the 46 are UAT test accounts); D closed, with the reason in the history (paid up, foreclosure, lapsed); F seen on the accounts whose balance was re-booked to another account (The Food Empire Limited, moved from MAIIC Agricultural to FInES in

December 2025 and to a term loan in May 2026, each move a closure credit matched by an equal debit on the next account).

Three accounts the batch did not settle cleanly. Imani Investments (000104460000061): the transfer, 1,102,536.37, exceeded what was owed, and the account is **in credit by 861,723.83**. Tac-Maz Investment (000104450000038): 37,291.56 is still owed after the transfer and interest is still charged monthly. Perisha Agro and Packaging (000104450000030): not in the batch; 1,513.12 still owed.

The write-off account over time.

When	GL 1064 movement	What it is	Loans
31 August 2024	22,286,434.32	Two write-offs posted as type 300	Micholess Creameries, Maluso Cooperative
January 2026	138,135,680.80	Debit 1064, credit 1876 ECL Write Offs: a memorandum pair; **no loan account moved**	Not identified (asked of Finance, section 5)
29 June 2026	38,492,032.54	TRF 3, type 305 on the loans, debit 1064	The 20 H accounts above

The write-off tables of E-Banker (LOAN_WRITEOFF_MASTER, LOAN_WRITEOFF_TRANS) are empty (CT_01), so the GL and the ledger are the only record. Of the three, only the 2024 write-offs carry the write-off transaction type.

The rule the engine follows. A credit to a loan account is a **write-off, not a collection**, when any of these holds: its transaction type is 300; it sits in a batch whose other leg is GL 1064 (read from `GL_05`, below, and later from the monthly ledger); or it is narrated "ACCOUNT CLOSURE" in a TRF batch on an account whose status is H. The loan book build (method B) records it as written off; the LGD engine counts the amount as a loss, not a recovery, and does not treat the account as settled; the EIR stops at the write-off date with the written-off amount as derecognition, not cash received; the RBM return shows it as written off. An account in credit after a write-off (Imani) is reported as a liability to the customer, not as negative exposure. Status H is mapped to written off in the engine's status table, and D "Paid Up" is not taken as evidence of repayment unless the balance went to nil by cash.

What is asked for. `GL_05` (pack 4, RUN 7): every leg of batch TRF 3 of 29 June 2026, and every posting to 1064 and 1876 in January 2026, in the same shape as `GL_03`. Finance: the confirmation that H means written off, the loans behind the January 2026 entry of 138,135,680.80, and what is to be done about Imani's credit and Tac-Maz's residual.

3.9 Reschedules: rebuilt from E-Banker's own tables (added 12 October 2026)

In plain language. Restructured loans are in version-1 scope (a negotiated change can be a modification under IFRS 9 5.4.3), but E-Banker's reschedule tables are empty (CT_01: LOAN_RESCHEDULE_DETAILS and T_RESCHEDULE hold no rows), so the plan was to wait for the vendor's Reschedule Report. The data dictionary shows that E-Banker keeps the evidence elsewhere, so the history can be rebuilt from the database instead.

What the stored loan books show, and why it is not enough. Between December 2024 and September 2026, 65 of the 144 accounts in the stored Loan Book Report change a term from one month to the next, but almost all of it is noise: in December 2025 the value date and the maturity change together (Lenziemill Milling's value date moves from 30 October 2020 to 1 September 2020), which is a correction of the report, not a reschedule, and some maturities flip between runs and back. Requiring the value date to stay the same and the change to last leaves four candidates (Nature's Gift twice, Ifracon, House of Men), three of them in that December. The loan books cannot stand in for a reschedule register.

Where the evidence is.

Table	Rows	What it shows
FIXLOAN_INST_CHART, rows with DELETE_FLAG Y or ACTIVE_FLAG N	part of 95,609	Superseded instalment rows: earlier versions of a schedule, with the date and the user that replaced them. The 6 October extract (P2_10) took live rows only
FIXLOAN_INST_CHART_300824, FIXLOAN_INST_CHART_141124, FIX_INST_CHART28SEP, FIXLOAN_INST_CHART1	4,977; 9,339; 9,106; 3,211	Copies of the instalment chart taken on those dates; a loan whose schedule differs from today's was rescheduled in between
MODIFICATION_LOG	5,991	Every edit: the table, the field, the old value and the new value, the date, who entered it and who passed it
ACCOUNT_CLOSER_HISTORY	21	The closure records of the 21 accounts with status H (section 3.8)

The method. RS_02a to RS_02g (pack 4, RUN 8) bring these in. A loan is taken as rescheduled when its schedule was replaced (a superseded set of instalments, or a dated copy that differs from the live chart in dates, amounts or the number of instalments) or when MODIFICATION_LOG records an edit to its maturity, tenor, moratorium, instalment or rate. Each reschedule is written with its date, the old and the new terms, and who made it, and feeds the reset-or-modification test of rate_change_classification (O17) and the derecognition threshold (O16). The vendor's Reschedule Report is held, not cancelled: it is asked for again only if RS_02 leaves gaps.

Two probes in the same pack. TRIAL_BALANCE_HISTORY (346,261 rows) is E-Banker's own trial balance history; if it holds every month-end, the monthly trial balance can come with the pack instead of separately from Finance. CUMOVUCH_OLD (13,359 rows) is an older copy of the ledger that may hold postings from before the July 2024 migration, the history the take-on loans lack (section 6.9). TB_02 and CV_01 (RUN 9) count what each holds before either is asked for in full, and TB_03 (RUN 9c) brings the trial balance rows for December 2025, August 2026 and September 2026: December and August are proved against the trial balances Finance supplied, to the cent (December also shows how the year-end closing is held), and if they tie, September's rows are the September trial balance and the monthly trial balance joins the pack (section 7.7).

3.10 Where the revenue roll-forward starts, and facilities drawn in tranches (added 13 October 2026)

In plain language. The engine builds each loan's amortised cost month by month: the opening balance, plus the interest at the EIR, less the cash. Everything after the first month follows from the month before, so the first month has to be right. Two faults were found in it on 13 October 2026 and both are corrected.

The first month was dated a month late. The opening was the present value of the remaining instalments taken at the end of the first month, and the month's interest was then added on top, so that month's interest was counted twice. A loan with one repayment of 1,030 due at the end of March, run from January at 1 percent a month, was left with about 10 outstanding after it was repaid in full. The opening is now the present value at the end of the month before (decision D32).

A facility drawn in tranches was treated as if it had been lent in full on day one. The schedule assumes every tranche is advanced at the start, so its present value counts money not yet lent, and the roll-forward never added a later tranche. Proven on Mphunzitsi SACCO (104430000085): approved 1,047.7 million, drawn 397.7 million in August 2025, 350 million in November 2025 and 300 million in February 2026; the engine opened December 2025 at 887.5 million against a ledger of 421.2 million and closed August 2026 at 349.9 million against 133.6 million. Malawi Police SACCO (3,139.3 million drawn on 19 January 2026) and KWACHA Boutique Hotel (800 million on 20 May and 700 million on 2 June 2026) were lent within their first engine month and were opened at the full present value with a full month's interest.

The rule now (decision D33). The first month opens on one of two bases, recorded on every row as opening_basis and carried forward with the closing balance:

Basis	When	Opening	Later advances
SCHEDULE	The usual case	Present value, at the locked EIR, of the contractual flows due after the end of the month before	Never added: the schedule's value already counts every tranche
LEDGER	The facility is still drawing down (an advance falls after the end of the month before), or was lent within the month (opening nil), or the borrower has prepaid (the ledger is below half the present value, decision D34)	The drawn balance in the E-Banker ledger at the end of the month before	Each tranche added in the month it is drawn, earning interest from that day; held on the row as `advances`

Every ledger opening is written to the audit log (`EIR Opening From Ledger`) with the present value, the ledger balance and the reason. The row now ties as opening + advances + interest + unwind - cash - derecognised = closing.

What it did to the figures (demo database, August 2026), first step. The ten loans on the ledger basis tie to the loan book within 2.8 million on 6,675.7 million: Mphunzitsi 127.2 million against 133.6 million, Malawi Police SACCO 3,065.1 million against 3,054.2 million. EIR interest for January to August 2026 read 1,812.3 million against contractual interest of 2,031.4 million, a shift of -219.1 million (it read -51.6 million before, because the double-counted first months raised the EIR side).

The headline number changed, and two further causes were found. After that step the other 21 loans, all on the schedule basis, were 479.5 million below the ledger by August 2026 and the gap grew every month. The largest were Chipala Investment (-117.6 million), Saile Financial Services (-70.1 million) and Chikomanyama Farms (-41.2 million). Until both causes were built, that part of the shift was overstated. Both are now built (13 October 2026, Edward's instruction to build it):

1. **Floating-rate loans whose rate has moved (decision D38, IFRS 9 B5.4.5).** The version 1 schedules of the floating-rate loans were generated at the contractual rate in the contract master on the day they were built, which is today's rate, not the rate the loan was lent at: Chipala Investment was lent at 30.0 percent and charged 29.1 to 30.2 percent through 2025, but its schedule was built at 26.1 percent, its August 2026 rate, and the EIR locked on it is 29.43 percent effective (26.07 nominal). The roll-forward therefore accrued about four points less than E-Banker charged for most of the period. The EIR in force for a month is now the locked nominal EIR plus the change from the rate the schedule was built on (`contract_eir.contractual_rate`) to the rate E-Banker charged that month (the loan book's rate, section 3.2), converted to an effective rate; the margin the fees add is kept. A fixed-rate loan keeps its locked EIR. Each row records the EIR it accrued at (`eir_annual`), and the GL reconciliation's rate effect and the EIR as at a date read it.
2. **Arrears at the opening (decision D39, section 6.9).** The present value at the end of the month before assumes every instalment up to then was paid as scheduled. Where the ledger owes more than the balance the schedule expects at that date (the same flows at the contractual rate) by more than one percent, the difference is booked as arrears and added to the opening, undiscounted because it is owed now. The first row records it (`opening_arrears`) and the audit log records the present value, the scheduled balance, the ledger balance and the arrears. Chipala had paid 152.2 million of 219.5 million due by 30 November 2025; its arrears at the opening are 81.3 million. In total 363.1 million of arrears were added across the book.

Where the figures stand now (demo database, 31 August 2026). Amortised cost 11,472.0 million against a gross ledger balance of 11,466.2 million, a difference of 5.8 million: the ten ledger-basis loans within 9.0 million, the 21 schedule-basis loans within 3.2 million (Chipala 907.9 million against 916.0 million). EIR interest for January to August 2026 is 1,913.7 million against contractual interest of 2,031.4 million: **the revenue shift is -117.7 million**. It was -51.6 million when the first months were double-counted and -219.1 million after the first correction alone.

4. Decisions, and the Governance Centre

4.1 Agreed this week

#	Decision	By	Why
D20	**The E-Banker record is the contractual record** (O19). The engine reads terms, rates, fees and cash flows from E-Banker. The signed offer letter is evidence; a difference between letter and system is reported to Credit as a control exception, never booked as a modification.	Dupleix, 7 Oct 2026; Dr Thom to confirm in writing	The signed terms were never keyed into E-Banker (F14), while the system now carries fees, margins and cash flows in full. It is the only record the engine can read completely. The twelve offer letters requested from Credit become test evidence, not a data source.
D21	**Every open choice of version 3 is a governed setting** with Dupleix's recommendation seeded as the value in force from 1 January 2025. Dr Thom confirms or changes each one in the Governance Centre when he is ready, with maker-checker and an effective date, and the engine records which value every month was run under.	Dupleix, 7 Oct 2026, at Edward's request	Nothing waits on a meeting. A choice that is changed later applies forward only; a month already run keeps the settings it was run under.
D22	**E-Banker is ingested through a landing zone and the monthly loan book is derived from the ledger** (section 6): raw tables that mirror E-Banker, loaded append-only by whichever of five routes MAIIC uses (three built, section 6.5), with gates that refuse a pack that does not tie, and one re-runnable build by any of three methods, the bootstrap of the stored report, the derivation from the ledger, or the printed report importer MAIIC uses today, chosen at any time.	Dupleix, 7 Oct 2026; the landing zone and the derivation added the same day at Edward's direction, the bootstrap and the report importer kept as options	The ledger is the primary record and ties to every other table; the report starts only in December 2024 and stores every re-run. Every derived figure traces to raw rows an auditor can open.
D24	**The Dupleix-suite layout is adopted for the user interface** (section 11): the six working groups with their colours, the icon rail, the page header with breadcrumb, the period chip, and light and dark mode as a per-user setting, as the Dupleix suite builds them. Routes, permissions and calculations are unchanged; screens move to where the suite puts them.	Edward, 7 Oct 2026	One shape of screen across every Dupleix system; the EIR screens of P8 are placed in it from the start
D25	**Dupleix's compliance audit workbooks are adopted** (section 12): five workbooks (IFRS 9 EIR, IFRS 9 impairment, IFRS 7 and IAS 1, RBM classification, Contract Schedule 1), each row naming the governance setting that governs it and the test that proves it; a register in the Governance Centre where MAIIC signs; an auditor's pack per period in the Report Hub.	Edward, 7 Oct 2026	Deloitte walks from paragraph to screen to test; the status counts are the project's status in one line

#	Decision	By	Why
D26	**One-command bootstrap from committed inputs** (section 6.10): the E-Banker pack, the data-dictionary results, the trial balances, the take-on workbook and the queries are committed under `docs/bootstrap/` with a manifest; `eir:bootstrap` builds a clean install, runs every engine in dependency order and verifies the result against the baselines and MAIIC's golden numbers.	Edward, 7 Oct 2026	Any server, including UAT and Deloitte's copy, is reproducible from nothing; the bootstrap is also the end-to-end test of the data foundation
D27	**Macro statistics are ingested from the World Bank and the IMF the way the suite does it** (section 13): indicator codes on the series, a fetcher and a parser, preview before commit, a batch of provenance on every observation, a command for the bootstrap and the scheduler, the Reserve Bank series by file, and a governed rule for which source wins. The existing tables and the six forward-looking screens are kept.	Edward, 7 Oct 2026	Forward-looking information with its source, address and time against every figure, which is what B5.5.49 to B5.5.54 and Deloitte ask for
D28	**The forward-looking adjustment gains a correlation finder, a repaired regression, a manual overlay route and lineage on the loan** (section 14); the arithmetic that applies the adjustment to PDs and feeds the ECL is unchanged.	Edward, 7 Oct 2026	Which series explains MAIIC's losses is found, tested and approved by two people; judgement has a governed road; every post-FLI PD says where it came from
D29	**The scenario set is a governed object per period and the ECL is weighted across scenarios** (section 15): three or more scenarios with narratives, weights, source vintage, calibration to Malawi's own history, shocks on the base path, two approvals, a lock, back-tests and sensitivity; the weighting moves from the macro path to the loss.	Edward, 7 Oct 2026	IFRS 9 B5.5.42 asks for the probability-weighted loss over a range of outcomes; IFRS 7.35G asks for the disclosure; the auditors ask for the governance
D30	**The Mega Farm programme loans stay outside the EIR engine and go fully into the ECL module** (section 16): they are Government money that MAIIC runs on its behalf, at 15 percent of which 5 is MAIIC's and 10 the fund's; 95 percent are in default and the losses are the fund's. The engine works out MAIIC's 5 percent share on the amount expected to be recovered, the ECL module stages and provides for the loans against the fund, and the Report Hub produces the Mega Farms notes of the financial statements.	Dupleix, 8 Oct 2026, on the 2025 financial statements; Dr Thom to confirm (O22)	The statements present the programme ring-fenced on every page; an EIR on a programme split would restate nothing they report, and the revenue-shift number is about MAIIC's own lending

#	Decision	By	Why
D31	**The staging thresholds follow the RBM DFI directive and reproduce the 2025 model** (section 3.6): Stage 3 at 181 days past due for medium- and long-term facilities, 91 for short-term and the Mega Farm programme; Stage 2 at 31 days; the 90-day presumption rebutted in writing on the directive; the four-consecutive-payments trigger and the days-past-due basis governed beside them; the 2026 policy note to say so.	Dupleix, 8 Oct 2026; Dr Thom to sign the rebuttal (O13)	The policy note says 90 days, the model applied 181 and the statements rest on it; the gap is closed by documenting the rebuttal, not by moving the number
D32	**The revenue roll-forward's first month opens on the amortised cost at the end of the month before** (section 3.10): the present value at that date of the flows still to come.	Dupleix, 13 Oct 2026	Taking it at the end of the month itself counted that month's interest twice.
D33	**A facility still drawing down, or lent within the month, runs on the ledger's drawn balance** (section 3.10): the opening is the drawn balance (nil when lent in the month) and each later tranche is added in the month it is drawn, with interest from that day; a schedule-basis loan never adds a tranche. The basis is recorded on every row.	Dupleix, 13 Oct 2026	The schedule assumes every tranche on day one; Mphunzitsi SACCO was overstated by 216 million at August 2026.
D34	**A prepaid loan opens on the ledger balance** when the ledger at the end of the month before is below half the present value of the remaining schedule; the substitution is audit-logged with both figures.	Dupleix, 13 Oct 2026	Barnet Agriculture repaid 55.6 million in November 2025; the schedule's value was 38.8 million against 0.4 million owed.
D35	**The actual cash of every loan is loaded from the E-Banker ledger** (`eir:load-ledger-cash`, source `EBANKER_LEDGER` in `eir_actual_transactions`) and is part of the month-end job, so the roll-forward uses receipts, not the schedule, wherever the ledger covers the month. A customer who has paid more than the ledger says is owed is shown as a credit balance, not a negative loan.	Dupleix, 13 Oct 2026	EIR as at a date read the schedule where the ledger held the receipts.
D36	**Every client is linked to its loans** (`clients:link-loan-book`, by external identity and then by name, recorded on the client) and **each import screen shows only its own import history** (`imports.kind`).	Edward, 13 Oct 2026	Clients were not linked to the loan book; every import screen listed every import.
D37	**Maintenance is seven jobs, as in the Stress Testing suite** (`maiic:task`): month-end (link clients, stage, PD by segment, forward-looking month-end, ledger cash, revenue run, Mega Farm ECL where present), refresh data, health check, update and bootstrap behind a typed confirmation, with a run history.	Edward, 13 Oct 2026	The tabs were overwhelming; the month-end order is now one button.

#	Decision	By	Why
D38	**A floating-rate loan's EIR is re-estimated each month with the rate E-Banker charged** (section 3.10, IFRS 9 B5.4.5): the locked nominal EIR plus the change from the rate the schedule was built on to the month's loan-book rate; a fixed-rate loan keeps its locked EIR; each row records the EIR it accrued at.	Edward, 13 Oct 2026	The floating loans' schedules were generated at today's rate, not the rate lent at; the roll-forward ran 479.5 million below the ledger with D39.
D39	**Arrears at the opening are booked as arrears** (sections 3.10 and 6.9): where the ledger at the end of the month before owes more than the balance the schedule expects, by more than one percent, the difference is added to the opening, undiscounted, recorded on the row and in the audit log.	Edward, 13 Oct 2026	The opening assumed every instalment had been paid; Chipala had paid 152.2 million of 219.5 million due.
D23	**The three MAIIC extract scripts are retired.** The engine's importers read the tables directly, through CSVs produced with the session settings of the 6 October request (ISO dates, point decimal).	Dupleix, 7 Oct 2026	Removes the three script faults of section 2.1 at source.

4.2 Decided and seeded: every setting, its default, and where it stands

The Governance Centre now holds 28 settings. The first 13 were there from P1 and P4; the 15 that follow were added on 7 October. "Status" says whether the value is agreed, decided on evidence, or Dupleix's recommendation awaiting Dr Thom.

Setting	Value in force (seeded default)	Status	Item
PLR change inside a month	**Whole month at the month-end rate (E-Banker)**	Decided on evidence, 7 Oct (section 3.1)	O2
When a floating loan's EIR is re-solved	At the PLR effective date	Agreed D8	
How a "Both" moratorium compounds	Monthly at posting	Agreed D10	
Which source decides whether a loan reprices	Interest Policy, then observed, then product family	Recommendation; the Interest Policy is now supplied	O3
Where the spread over prime comes from	LIVE_AT_DRAWDOWN	Recommendation; the evidence of 3.2 supports it	O1
Interest on Stage 3 loans	Net carrying amount	Agreed D11	
Day count	ACT/365	Agreed D9	
Rate for a quarterly or annual period	Monthly compounded (E-Banker)	Proven 29 Sep	
Where actual cash received comes from	Loan book (change in Repayments)	Agreed D14; the ledger now gives the same figure exactly	
Loans with Interest Policy = Manual	Read the rate monthly, each change is a reset	Recommendation	O4
The derecognition test	10 percent	Recommendation	O16
When a difference is an exception	1 percent of the posted amount, floor MWK 1	Recommendation, with Deloitte	O11
A Repayments counter that falls	Discontinuity: take cash from Extract B (now the ledger)	Recommendation; the ledger makes it exact	O5
Which record is the contractual one	**E-Banker system record governs**	Decided D20	O19

Setting	Value in force (seeded default)	Status	Item
Interest basis on a partly drawn facility	Per-account flag, then the scheme, else refuse	Recommendation; the flags are now supplied	O6
Which schedule is the expected cash flow	Core dates and rate; LOS schedule is reference only	Recommendation; follows D20	O7
Loan books before December 2025	Stored loan book history, every month-end	Decided D22; superseded by the build-method setting below, which covers every month	O8
How E-Banker data arrives: the feed route (`ebanker_feed_route`)	Route 1, manual pack	Decided D22: routes 1 to 3 of section 6.5 are built (13 Oct 2026), routes 4 and 5 are not; MAIIC switches the route in force at any time; routes 4 and 5 need Dr Thom and ICT (and route 5 the vendor)	O9
Pre-migration history of the take-on loans (`takeon_history_basis`)	Recompute from origination where the block and fees exist, else start at the take-on balance	Recommendation (section 6.9); the loan carries the basis it was built on	new
Which macro source wins where two overlap (`macro_source_precedence`)	World Bank for actuals, IMF WEO for forecasts, RBM file for rates; manual overrides only with a reason	Recommendation (section 13.5)	new
How the forward-looking adjustment is produced (`fli_adjustment_route`)	Regression	Recommendation (section 14.6); the manual overlay and the combined route are the alternatives	new
How the adjustment reaches the PD (`fli_transmission_method`)	Multiplicative scalar on the proxy ratio	Recommendation (section 14.7); the segment scalar, the nine reference methods, the logit model and the Vasicek shift are the alternatives, each declined until its data exists	new
Asset correlation for the Vasicek shift (`fli_asset_correlation`)	0.12 per portfolio	A starting value in the range regulators use for corporate exposures; used only under the Vasicek method	new
Expected-sign test on a regression pair (`fli_expected_sign_test`)	Required	Recommendation (section 14.4)	new
R-squared cut-off for an approvable model (`fli_r2_cutoff`)	30 percent	Recommendation (section 14.4); MAIIC may tighten	new
Minimum observations for a fit (`fli_min_observations`)	12	Recommendation (section 14.4, the guardrail)	new
Significance level for a fit (`fli_alpha`)	5 percent	Recommendation (section 14.4, the guardrail)	new
Normality limits for a series profile (`fli_normality_limits`)	Skewness 1.0; excess kurtosis 3.0	Recommendation (section 14.4, the profiler)	new
How scenarios are weighted (`scenario_weighting_method`)	Weight the ECL across scenarios	Recommendation (section 15.5); the macro-path method kept for reconciliation	new
Minimum scenarios in a set (`scenario_minimum_count`)	3	Recommendation (section 15.6)	new
Floor on the base weight and ceiling on any weight (`scenario_weight_bounds`)	Base at least 40 percent; none above 60 percent	Recommendation (section 15.6)	new
Calibration note required on a downside (`scenario_calibration_note`)	Required	Recommendation (section 15.6)	new

Setting	Value in force (seeded default)	Status	Item
An overlay needs an approved scenario set (`overlay_requires_approved_set`)	Required	Recommendation (section 15.7)	new
How the loan book is built (`loan_book_build_method`)	Method B, derived from the ledger	Decided D22: all three methods of section 6.2 are built (bootstrap of the stored run, derivation from the ledger, the printed report importer); MAIIC switches at any time; the method used is recorded on every row	O8
Which GL absorbs the EIR true-up	Dedicated EIR adjustment income account	Recommendation; Finance opens the account. Today the true-up lands in 4206 Interest on Term Loans-MAIIC (section 3.7), which is the fallback until the new account exists	O10
Shape of the auditor export	Summary-tab shape	Recommendation, with Deloitte	O12
Keyman insurance charged to the borrower	Not integral: pass-through, outside the EIR	Recommendation; a Phase 0 sign-off	O13
Nascomex preference shares	Equity instrument (IAS 32), outside the EIR	Recommendation; a Phase 0 sign-off	O13
Rebutting the 30-day Stage 2 presumption	Allowed with documented evidence, approved	Recommendation; a Phase 0 sign-off. The 90-day default presumption is already rebutted on the RBM DFI directive of 2018 (section 3.6, D31): Stage 3 at 181 days for medium- and long-term facilities, 91 for short-term and Mega Farm; the policy note of 2026 to record it	O13
Days-past-due basis (`dpd_basis`)	Oldest overdue instalment (the directive's count)	Recommendation (section 3.6); bucket ageing shown beside it	new
Missed-instalment trigger for Stage 3 (`stage3_missed_instalments`)	4 consecutive	The accounting policy's own second test (note 22.8.1); a Stage 3 trigger beside the day count	new
Reset or modification	In-contract moves reset; negotiated changes modify	Recommendation; Deloitte in writing before the first reset is booked	O17
Approving a version 1 schedule	Second person must approve	Recommendation	O18
The historic materiality assessment	Reproduce it beside the detailed figure	Recommendation; Finance supplies the threshold	O20
Which source a contract's fees come from (`fee_source_precedence`)	GL posting, then Finance workbook, then LOS capture, then Extract A	To build (P4b, section 7.8.2); a lower source is superseded, never added	new
Fee imports checked against the fee GLs (`fee_gl_reconciliation`)	On: check fee imports against 4871 and 4873	Built 11 October 2026 (section 7.4); reports and never blocks; Off stops the check and the screens say so	new
The year-end fee reclassification journal	Engine journal replaces the manual reclass	Recommendation; follows O10. The 2025 move (259,317,000 from 4873 into 4206) was a manual adjustment recommended by the auditors from an extrapolated sample, applied in the TB-to-AFS mapping and not posted (section 3.7); the engine's proposal is reconciled to it	O21
Mega Farms facilities	**Out of the EIR engine; in the ECL module; MAIIC's 5 percent share on the net carrying amount**	Decided D30 on the 2025 financial statements (section 16); Dr Thom to confirm	O22

Setting	Value in force (seeded default)	Status	Item
Probability of default for the Mega Farm schemes (`megafarm_pd_method`)	MAIIC agricultural-sector PD scaled to the programme	Recommendation (section 16.8); the seasonal cohort rate is the check; the programme's own transition matrix is declined until three seasons exist; the method used for the 2025 provision to be confirmed by Finance	new
Programme scalar ceiling for the Mega Farm PD (`megafarm_scalar_ceiling`)	40 times the sector PD	A scalar above it needs a second approval (section 16.8); the 2025 cohort implies about 30	new

Seven of these need a genuine choice from Dr Thom rather than a confirmation: the true-up account (O10), the derecognition threshold (O16), reset or modification (O17), the three Phase 0 sign-offs (O13), the fee journal (O21) and Mega Farms (O22). The rest are what the data shows.

Two things are not settings and are tracked elsewhere: the materiality threshold *amount* (O20), which becomes a governed amount once Finance gives the figure; and the fourteen open items that are facts for people to supply (section 5).

5. What is still owed by people at MAIIC

Nothing further is needed from the database to start building. These are the answers only people can give, with the document that asks for each.

Who	What	Asked in	Why it matters
Tamanda	Tick the take-on mapping workbook (4 unmatched, 11 carrying-amount differences above 5 percent)	`Take-on schedules with mapping - for Tamanda to confirm - 7 Oct 2026.xlsx`	Loads the 109 take-on loans
Tamanda	The fees of the take-on loans, in the yellow columns of the returned workbook (O14), and the explanation of the 2024 arrangement fee of MWK 1.30 billion in the ledger (O23, restated in section 3.7); and whether the BRFF fees of June 2026 belong to a programme outside the eighteen schemes, so that the extracts cover it	The workbook with fee columns of 7 Oct; the 2024 question in version 3; section 3.7	Fees are the EIR; the BRFF answer decides the scheme list in the queries
Credit	The twelve offer letters: the ten samples and the two over-sanction accounts (Mchinji 50m against 100m drawn; VNC Bricks 20m against 200m); the 31 active accounts with no security on the register (`P2_11`)	`Request to Credit - the twelve offer letters - 7 Oct 2026.pdf`; the collateral list of 8 Oct	Test evidence under D20; the two sanction checks are a control finding; the loss-given-default rests on the register
Finance	Which 2024 figure agreed to the audited accounts for 1050201 and 1050202, then the GL opening adjustment	`Outcome - GL differences and Zaithwa Farms explained - 7 Oct 2026.pdf`	The ledger-to-accounts bridge

Who	What	Asked in	Why it matters
Finance	Whether the year-end re-rating of 31 December 2025 was intended as policy (the posting itself is now fully explained, section 3.4); which 25 loans the November 2025 ECL model leaves out of E-Banker's 125 and why (section 3.6); the historic materiality threshold; **how the 2025 Mega Farm provision of K39.76 billion was arrived at** (the staging rule, the probability and loss assumptions, the treatment of maize and the buyer receivables)	`Note to Barry - the 28 interest adjustments of 31 Dec 2025.pdf`; O20; section 16.8	December 2025 interest; the historic assessment; the starting point for the Mega Farm provisioning method
Finance	Confirmation that status H means written off and closed; the loans behind the January 2026 write-off of 138,135,680.80 (1064 against 1876, no loan account moved); what is to be done about Imani Investments' credit of 861,723.83 and Tac-Maz's residual of 37,291.56 (section 3.8)	Pack 4 email; section 3.8	Write-offs are losses, not collections: LGD, the EIR and the RBM return
Barry / vendor	`GL_05`, every leg of batch TRF 3 of 29 June 2026 and the January 2026 postings to 1064 and 1876 (section 3.8); the Zaithwa Farms balance rows rebuilt; the reschedule history through `RS_02` (pack 4 RUN 8, section 3.9; the vendor Reschedule Report is held unless `RS_02` leaves gaps) the two probes `TB_02` and `CV_01`, and the trial balance rows `TB_03`; the rest of pack 2 (the 30 September loan-book run, the eight Mega Farm files) and the `P2_13` re-run with the payload rendered as text (`DBMS_LOB.SUBSTR`)	The notes of 7 Oct; pack 2 of 8 Oct; version 3 phase P7	Which accounts are live; restructures are version-1 scope; the Mega Farm data; the fees and rates at approval
Dr Thom	Confirmation of the Mega Farm treatment (D30, section 16), the written confirmation of D20, and the remaining choices of section 4.2	This document; the O22 paper of 8 Oct	The headline number

6. Ingesting E-Banker: the landing zone and the feed

6.1 In plain language

Until now the system has been fed by hand: a report printed from E-Banker, saved through Excel, uploaded once a month. Version 3 proposed loading the stored Loan Book Report table instead. The follow-up extracts showed something better is possible: the ledger itself, the primary record of every posting, ties to every other table we hold, so the monthly loan book can be **derived from the ledger** rather than copied from a report. And because the extracts are now ordinary queries with known keys, they can arrive by whichever road MAIIC finds convenient, from a file Barry uploads to a scheduled read over the VPN, without changing anything downstream.

The design has three parts. A **landing zone**: raw tables that mirror E-Banker's, loaded exactly as received and never edited. A **build** of the monthly loan book from the landing zone by one of three methods, all kept and all available at any time: the **bootstrap**, which takes the stored Loan Book Report's latest run for each month as it is; the **derivation**, which builds the row from the ledger; and the **report importer**, the existing upload of the printed Loan Book Report as an Excel file, which MAIIC has used every month until now; one command, re-runnable whenever a rule or a source row changes. And a **feed**: the one door through which every pack of

extracts enters, whichever road it came by, with gates that refuse a pack that does not tie.

6.2 The three build methods

Method A, the bootstrap. The loan book row is the stored Loan Book Report's latest run for that account and month-end (`ebanker_loan_book_runs`, highest row id per account-month), column for column: E-Banker's own carrying amount, principal, interest to date, repayments, arrears, status and rate, mapped once to `loan_books` through a saved template. It is the simplest method, it is what MAIIC's own report says, and it is already tied to the ledger on 2,264 of 2,264 account-months. It covers December 2024 onward, which is where the stored report begins; the months from the take-on at 31 July 2024 to November 2024 are built from the balance history and the ledger and marked as derived.

Method B, the derivation. For every account and every month-end from the take-on to the latest month, the row is built from the primary records as follows.

Loan book field	Source	Why
Carrying amount, principal, interest to date, cumulative repayments, disbursed	The ledger's running balance and its postings by type	The primary record; proven against the balance history (2,533 of 2,549) and the stored loan book (2,264 of 2,264)
Approved amount, value and maturity dates, tenor, product, GL, customer	The loan master and account master	The record of the facility
Rate	The rate charged, from the stored loan book's rate column, with the PLR and the rate set-up as the register of changes (section 3.2)	The rate actually posted
Arrears buckets, overdue days, status, segment	The stored loan book's latest run for that month-end	E-Banker's own arrears arithmetic depends on its instalment chart, which is unreliable to reproduce, and these are the figures MAIIC reports to RBM
Undisbursed commitment	Approved less disbursed, checked against the disbursement schedule	Both sources held

Under method B every month from July 2024 is built the same way, and beside every derived carrying amount the stored loan book's figure is kept; a difference is a flagged row, never a silent choice.

Method C, the report importer. The printed Loan Book Report (Menu ID 3868), uploaded as the Excel file MAIIC produces today, through the importer that already exists (`ProcessLoanImportJob`), unchanged in what it reads. Two things are added so that it lives inside the same design: the parsed rows are landed in `ebanker_loan_book_reports` with the file's hash and load id before `loan_books` is written, so a month loaded this way traces to its file like any other; and the file passes the date gate of 6.4 (every date unambiguous, the row named on failure), because this is the route on which Excel scrambled dates in September. It needs no query and no access to the database, which is its value: a month can be loaded from the report alone when no pack exists, by whoever has the report. It gives the report's figures, like method A, for the months the report was run.

Choosing between them. The method in force is a Governance Centre setting (`loan_book_build_method`, section 4.2) with the three methods as its options, changed under maker-checker with an effective date; a month is built by the method in force on its period end, and the method used is recorded on every row. All three read from the landing zone and pass its gates, so switching changes no route and no screen. Dupleix's recommendation is method B, because it covers every month from the take-on on one basis and traces every figure to a posting; method A is the right choice while the derivation is being proven, for a quick reload of a single month, or whenever MAIIC prefers the report's own figures to stand; method C is the right choice for a month with no pack, or if the feed is ever interrupted, since it needs nothing but the report. Whichever is in force, the others remain available on the feed screen for a named month.

6.3 The landing zone

Raw tables that mirror E-Banker column for column, under their own names, loaded append-only.

Table	Mirrors	Key	Loaded
ebanker_ledger	CUMVOUCH (P1_01)	CUMVOUCH_DET_ID	Incrementally by key, plus a re-pull of the last two months to catch back-dated postings and deletion flags
ebanker_balance_history	ACCOUNT_BALANCE (P2_09)	ACCOUNT_BAL_MST_ID	Incrementally by key
ebanker_loan_book_runs	LOAN_BOOK_DETAILS_ALL (P2_08)	LOAN_BOOK_DET_ID_A	Incrementally by key; every run kept, the latest per account-month used
ebanker_account_master, ebanker_loan_master, ebanker_rate_setup, ebanker_plr_master, ebanker_charges, ebanker_disbursement_schedule, ebanker_status_history	P1_02, P1_03, P1_04, P2_06, P2_07, P3_12, P3_13	Their own ids	Whole each time; they are small
ebanker_trial_balances	The monthly trial balance files as received from Finance (20 to date), the AFS bridge workbook and the mapping to the audited accounts	File hash and row number	One table per file; every GL line exactly as printed, with its cumulative year-to-date balance. gl_trial_balance_lines, the GL side of the reconciliation, is derived from it on build; the year-to-date rule of section 7.7 is applied on read, never stored
ebanker_loan_book_reports	The printed Loan Book Report as uploaded (method C)	File hash and row number	One table per upload; the parsed rows exactly as read
ebanker_loads	The packs themselves	Pack hash	One row per pack: period, route, who, when, the manifest, the gate results, the watermark after loading

Three rules. **Nothing is overwritten**: a source row that arrives again with different content (a back-dated change, a deleted flag set) is stored as a new version with its load date, so a month can be re-derived exactly as it looked at the time. **Every row carries its pack**: the hash of the file it came from and the load id, so every derived figure traces to raw rows an auditor can open. **The zone is read by the build only**: no screen edits it.

6.4 The pack: one contract for every route

A pack is a set of files, one per query, plus the month's trial balance from Finance, plus a manifest. The manifest (manifest.json) records the period, the run timestamp, the session settings used (ISO dates, point decimal, the RUN 0 settings of the 6 October request) and, for each file, the query id, the query version, the row count and the SHA-256 hash.

- **The queries are versioned in the system**, not in an email. Data Foundation, E-Banker Feed, Queries holds the SQL text of each extract with a version number and a checksum, and hands Barry the file to run. A column added later is a new version; the manifest says which version produced each file.
- **Watermarks.** For each incremental table the system records the last key loaded; the query for the next pack is "where the key is greater than the watermark", plus the two-month re-pull. Masters come whole.
- **Gates, run before anything is derived.** Dates parse as ISO throughout; numerics parse; counts and hashes match the manifest; the query versions are ones the system knows; every ledger account exists in the account master; the balance history ties to the ledger running balance at every month-end in the pack;

every in-scope account has a month-end row; the month's movement on each fee GL in GL_04a equals the trial balance's movement on that GL (section 3.7), since the account ledger alone does not carry every fee (this gate is new on 11 October 2026 and is built with GL_04 in P4b). A failure refuses the pack with the file and row named; the raw rows are kept in quarantine against the load; nothing downstream moves.

- **The fee control is a standing monthly query** (added 11 October 2026). GL_04a takes every posting to 4871, 4873, 4874 and 4875 since the watermark, and GL_04b every other leg of the same batches, which shows the loan account or bank side of each fee. Both are in the register and in pack 4; from the October month-end they join M09_01 to M09_03 in every monthly pack.
- **A large extract may come in parts.** Where SQL Developer cannot hold a result (the Mega Farm ledger, MF_03, stopped at 107 of about 3,685 accounts on 9 October), the run sheet splits it by account range into parts that each export from the grid. The parts are named MF_03_ledger_a.csv to MF_03_ledger_g.csv, so that both a hand-written manifest and the route 2 script (which reads the query id from the first two parts of the file name) put every part under MF_03; rows are keyed on CUMVOUCH_DET_ID, so the parts neither overlap nor need a merge. The counts come in CHECK_MF_03_row_counts.csv, a check file that is never landed.

6.5 The five routes

The printed report is not a route: it enters by method C with no pack, and is the one way in that needs no query. Every route produces the same pack and enters by the same door; the ingester does not know which road a pack came by, and moving from one route to another changes nothing downstream. Routes 1, 2 and 3 are built; routes 4 and 5 are specified but not built (status at 13 October 2026, below). **MAIIC chooses the route in force at any time**: it is a Governance Centre setting (ebanker_feed_route, section 4.2) with the five routes as its options, changed under maker-checker with an effective date like every other setting. The route in force is the one the scheduler runs; the others stay available, so a manual pack can always be loaded, for a missed month or a correction, even while the scheduled read is live. The order below is the order in which MAIIC can have them, not an order of preference.

Route	How the pack is produced and delivered	What it needs	Place
1. Manual pack	Barry runs the versioned queries in SQL Developer with the RUN 0 settings, zips the CSVs, uploads them on the E-Banker Feed screen; the screen builds the manifest from the files and runs the gates	Nothing new: it is what happened on 7 October, made repeatable	Now. Loads the history (section 6.7) and serves any month until route 2 is in place
2. Scheduled export at MAIIC	A SQLcl or SQL*Plus script Dupleix writes, parameterised by period and watermark, run by Windows Task Scheduler on a MAIIC server on the first working day after month-end; it writes the pack to an SFTP folder or a network share; the system's scheduler polls the folder and ingests	A read-only Oracle account for the script; one folder; Barry to install the task	The monthly mode. Weeks, not months
3. Push by API	The same script posts the pack to an endpoint on the system with an API token over HTTPS; no shared folder	A token, outbound HTTPS from the MAIIC server	The alternative to route 2 where a shared folder is awkward

Route	How the pack is produced and delivered	What it needs	Place
4. Direct read over the VPN	The system connects to E-Banker's Oracle database read-only (Laravel's OCI8 driver with the Oracle Instant Client), runs the versioned queries itself on schedule and writes straight into the landing zone; the pack and manifest are generated internally, so the audit trail is identical	Read-only credentials over the VPN already in place (the dictionary query DD_18 confirmed what a read-only account can see); the vendor's consent; Dr Thom's authorisation	The destination, and the answer to O9. When ICT agrees
5. Vendor view or API	Virtual Galaxy exposes a database view or an endpoint carrying the same columns; route 4 reads it	A paid change request to the vendor	Only if route 4 is refused

Status at 13 October 2026. Route 1 is built and in use: the packs of 7, 8, 9 and 12 October were landed by it (Customer & Loan Data, Imports & Feeds, E-Banker Loads). Route 2 is built (`eir:poll-feed-folder`, run daily at 06:30 by the scheduler) and waits for MAIIC's side: the export script on a MAIIC server and the shared folder. Route 3 is built (`POST /api/ebanker-feed/pack` with a bearer token; the poller lands what it receives) and waits for the token to be issued. Route 4 is not built: the route name exists in the code, but there is no Oracle connection, driver or query runner; it needs ICT to agree read-only credentials over the VPN and the Oracle Instant Client on the server. Route 5 is not built and is a paid change request to the vendor. Until ICT agrees route 4, route 1 serves every month and route 2 is the next step.

Finance's trial balance is not an E-Banker pack and has its own door: EIR & Revenue Recognition, EIR Data, **Trial Balances** (built 13 October 2026). Finance uploads the month's E-Banker trial balance (xls or xlsx); it is landed through the landing zone's gates (a known month, debits equal to credits, the report's grand total) with its hash, a file already landed is not loaded twice, a corrected file for a month replaces that month's lines and keeps the old version, and months loaded before the landing zone existed are left as they are. The command `eir:import-trial-balances` and the bootstrap use the same service.

Whichever route is in use, the system never writes to E-Banker; the Oracle role is read-only; credentials live in the environment file, not in code; and the extracts carry customer data, so route 2's folder and route 3's endpoint are restricted to the two systems and logged.

6.6 Building the loan book

`eir:build-loan-books {from} {to} {--method=bootstrap|derive|report}` builds `loan_books` for the months asked, from the landing zone, by the method in force unless one is named (6.2). It is idempotent on account and reporting period, re-runnable at any time, and run under maker-checker from the feed screen: one person asks for the build, a second approves it, and the audit log records the method and the pack hashes it read. Rebuilding a month already run, by any method, prints every difference first; a locked period is never restated. The ECL columns on `loan_books` (stage, LGD, forward-looking adjustments) are not touched by any of the three methods.

6.7 Loading the history, once

The history is loaded by the bootstrap of section 6.10 from the committed inputs; the steps below are what it does and how it is proven.

1. **The 7 October pack is committed** under `docs/bootstrap/ebanker-pack-2026-10-07/` with its manifest and hashes; it is pack 1 under route 1. The files are never opened in Excel.
2. **Gates.** The pack passes 6.4 with two accepted exceptions recorded against the load: the FInES GL openings of section 3.5 until Finance corrects them, and the Zaithwa Farms balance rows from May 2025 until the vendor rebuilds them.

3. **Build on a copy of the production database** for every month from July 2024 to August 2026, by method A first (the stored report is the quickest proof) and then by method B, and compare the two: the carrying amounts must agree on every account-month except the flagged rows of section 3.5. December 2025 to August 2026 are already loaded from the Excel reports; the build prints every difference before it overwrites.
4. **Prove.** Run one ECL month and one EIR month on the copy and compare with the Excel-loaded results; the Baselines sheet of section 12 must show PASS on every tie of section 9.
5. **Production**, then the same every month by route 1 until route 2 is installed. The first monthly pack, for September 2026, was run on 8 October and behaved as the contract requires: the August re-pull was identical to the first load (no back-dated posting), September arrived complete, and the month-end balances agreed with the ledger on every account.

A side benefit stands: with twenty-six months in `loan_books`, the ECL module can be back-run month by month.

6.8 The feed screen

Data Foundation, E-Banker Feed: the queries (versioned, downloadable as the file Barry runs); the load history (pack, route, period, the gate results or the refusal reason, who loaded, when); the watermarks; the quarantine; and the Build action (method A, B or C, a month or a range; method C takes the report file) with its approval. Every derived loan-book row links back to its raw rows and its pack.

6.9 The take-on schedules

What they are for. E-Banker's history of the 109 take-on loans begins on 31 July 2024 with one opening posting each. The effective interest rate needs what happened before that: the origination date, the original principal, the contractual rate and instalment pattern, and the fees charged at the start. Tamanda's workbook of amortisation schedules at 31 October 2024 (100 blocks, mapped on 7 October to 105 of the 109 facilities) is the only record of that. The engine uses it for two things: to solve the EIR at origination and roll the amortised cost forward to 31 July 2024, so that the loan enters E-Banker at its true amortised cost rather than its take-on balance; and as the version 1 schedule of those loans (`schedule_source = TAKEON_WORKBOOK`) in place of a generated one.

The fees. Neither the workbook nor E-Banker records the fees charged when a take-on loan was granted, and the EIR cannot be solved without them. The mapping workbook returned to Tamanda on 7 October (`Take-on schedules with mapping - with fee columns - for Tamanda to confirm - 7 Oct 2026.xlsx`) carries, on its Upload summary sheet, seven columns for her to complete per facility: the arrangement fee, the legal fees, any other fee that was a condition of the loan, the date charged, whether the fees were deducted from the amount paid out, the offer letter or receipt the figures come from, and a total. A blank means no such fee; a zero means known to be nil. This replaces the separate fee template of 25 September for the take-on loans (O14 is unchanged in substance: Finance supplies the fees; the system reads them).

Landed, not typed. The workbook enters by the feed door as a pack of its own kind: its hash, then two raw tables. `takeon_blocks` holds one row per block: title, principal, rate, term, start date, the mapped account, the mapping confidence, Tamanda's tick, the fees, and the sheet and cell each value came from. `takeon_schedule_lines` holds one row per instalment line: due date, instalment, principal, interest, balance, serial, with its cell reference. Every figure traces to the cell in the workbook she signed; the Upload summary's formulas already point there.

Gates. Every block is mapped to one account that exists in the master, or is explicitly marked not matched or refused (the three equity positions); no account has two blocks; the block principal equals the take-on posting (exact on 77) or the difference is named; the schedule's balance at 31 July 2024 is compared with E-Banker's take-on balance and the difference flagged as arrears or prepayment at take-on; every date is unambiguous; serial numbers out of due-date order are sorted by date and the anomaly raised (F16); a mapped facility with

no fee row is flagged, never assumed fee-free. A refusal names the block and the cell.

The build writes `contract_takeon` per account: origination date, original principal, contractual rate, fees, and the pre-migration cash flows. Those flows are contractual, not actual: nobody holds the receipts before E-Banker. How the engine treats them is a governed setting, `takeon_history_basis`:

Option	Meaning
Recompute from origination where the block and fees exist, else start at the take-on balance (seeded)	Where a facility has a mapped block and its fees, solve the EIR on the original schedule and fees, assume instalments were paid as scheduled to 31 July 2024 except where the take-on balance says otherwise, and book that difference as arrears at take-on. Where it has not (the four unmatched facilities, the eight closed accounts without a block, any block still without fees), treat the 31 July 2024 carrying amount as the opening amortised cost with no day-one history, and flag the loan
Recompute from origination for every take-on loan	As above, and refuse a loan whose block or fees are missing until they are supplied
Start every take-on loan at its take-on balance	No pre-migration history for any of them; the EIR is solved from 31 July 2024 on E-Banker's flows

The seeded option is the reasonable one: it uses the history wherever MAIIC can evidence it and is honest where it cannot, and every loan carries the basis it was built on so an auditor sees which. Changing the setting rebuilds the take-on population only.

Where. The workbook is committed under `docs/bootstrap/takeon/` and loaded by the bootstrap (6.10); the returned version with the fees replaces it as a new file. Data Foundation, Take-on Schedules: upload a workbook (the alternative to the committed copy), see each block with its mapping, confidence, tick and fees, the gate results, and the Build with approval. Tamanda may confirm a mapping or enter a fee in the screen instead of in Excel; the screen's entry is the one that counts and is audit-logged. Re-uploading creates a new version; the previous build is kept.

6.10 The bootstrap: a clean install that loads itself and proves itself

The Dupleix suite installs a client system with one command that wipes a clean database, seeds it, loads the client's own input files from a folder committed in the repository, runs the engines and checks the result against the golden numbers. MAIIC adopts the same method (decision D26), so that any server, including Deloitte's copy and the UAT copy, is built from nothing to a proven state without anyone sending a file.

The committed inputs. `docs/bootstrap/` in the repository holds, exactly as received and never re-saved: the E-Banker pack of 7 October 2026 (the 18 extracts and the 5 afternoon queries, 23 files), the data-dictionary results of 6 October (24 files), the 20 monthly trial balances of January 2025 to August 2026, the December 2025 AFS bridge in both received versions, and the mapping of every GL line to the audited accounts with the year-end balances of 2023 to 2025 and August 2026, the take-on workbook with its fee columns, the SQL that produced the pack, and `manifest.json` with every file's query id, row count and SHA-256 and the accepted exceptions the gates allow. The committed copy is read first; a OneDrive path is only a local fallback for development. A replacement, for example the workbook Finance returns with the fees filled in, is a new file beside the old one and a new manifest entry; nothing in the folder is edited in place. The files are MAIIC's data (borrower names and balances); the repository is private and access to it is governed as access to the production database is.

The command.

```
php artisan eir:bootstrap --fresh --with-client-inputs --build --run-engines --verify
```

Step	What it does	Idempotent
1	`migrate:fresh` on `--fresh`, refused if user data exists unless `--force-wipe` is also passed	Safety, not a step

Step	What it does	Idempotent
2	Seeds: roles and permissions, the Governance Centre defaults (section 4.2), the help centre, the fee rulebook, the compliance-audit modules (section 12)	Yes: a key that exists is left alone
3	Lands the committed pack and the trial balances into the landing zone by route 1 (section 6.4): the manifest is checked file by file, the gates run, the accepted exceptions are recorded against the load	Yes: a file whose hash is already loaded is skipped
4	Lands the take-on workbook (section 6.9) and the fees it carries	Yes
5	On `--build`: builds the loan books for every month from July 2024 to the last month in the pack by the method in force (section 6.2), builds the take-on population under its basis, generates the version 1 schedules	Yes: a locked period is never restated
6	On `--run-engines`: runs the engine chain of 6.10.1 end to end for every period from the first full month to the last month in the pack, in the order the modules depend on each other, synchronously (the queue driver is set to `sync` for the run, so no worker is needed and nothing is left waiting)	Yes: a period already run and locked is skipped
7	On `--verify`: runs the baselines of section 9 and the golden numbers of 6.10.2 against the database, prints the table, PASS or FAIL per row, and exits non-zero on any FAIL	Yes

6.10.1 The engine chain the bootstrap runs

The suite's bootstrap runs its engines in dependency order and treats the result as the proof of the installation. MAIIC's chain, in the order the modules depend on each other, with what each needs and how the bootstrap supplies it where a person would normally act:

Order	Engine	Entry point	Needs	How the bootstrap meets it
1	Macro statistics	`macro:import-worldbank`; the WEO snapshot	Internet, or the committed snapshot	Snapshot under `docs/bootstrap/macro/` when the fetch fails (13.3)
2	Scenario set	the seeded first set of 15.8	An approved set	Seeded as *proposed*; the bootstrap approves it under the automated label so the chain can run; it is not a MAIIC approval and the screen says so
3	Staging and SICR	`StagingClassifier` on each month's loan book	Governed thresholds; the loan book	Thresholds seeded; loan books built in step 5
4	PD: transition matrices	`TransitionMatrixService`, cumulative	Twelve or more months of graded loan books	The 26 months from the landing zone
5	LGD	`CalculateLGDJob` and its chunks	Payments and collateral	Payments from the ledger; collateral from the register; runs synchronously
6	Forward-looking adjustment	the route in force (14.6)	An approved model, or an overlay	With no approvable model on a clean install the bootstrap runs the **manual overlay route at zero** and marks every post-FLI PD "no adjustment: bootstrap"; the regression route is used once a model has been approved by two people
7	ECL	`ifrs9:recalculate-ecl --pd=pd_post_fli`, the time-phased service	PD, LGD, EAD, stage, the EIR for discounting	Steps 3 to 6 and 9

Order	Engine	Entry point	Needs	How the bootstrap meets it
8	EIR: schedules	`eir:generate-schedules`	Terms, drawdowns, the take-on basis	Steps of 6.9; version 1 approved under the automated label
9	EIR: solve and revenue	`CalculateEirJob`, `eir:run-revenue {period}`	Approved schedules, fees, rates, the governed conventions	Step 8; fees from the charges table and the take-on workbook
10	GL reconciliation	`EirGlReconciliationService`	The trial balances	Landed in step 3 of the bootstrap
11	Stress testing	`StressTestingController` logic as a service	The ECL of step 7	Run for the seeded scenario set
12	Reports and the audit workbooks	the 30 reports; `tools/compliance`	Everything above	Generated for the last period; the Baselines sheet reads the live figures

Two orderings matter and are enforced: the ECL is discounted at the EIR, so step 9 runs before step 7 for each period (the chain runs 8 and 9 first, then 3 to 7, then 10 to 12); and the forward-looking adjustment runs once per scenario under the seeded weighting method of 15.5, so step 6 is a loop over the set of step 2.

Three engines are queued jobs today (LGD, EIR solve, revenue). Under `--run-engines` they run synchronously; on a server they still run through the queue, and the installation guide carries the worker command with a memory ceiling, because the suite found that a worker left on its default limit stops itself part-way through a full chain and later jobs wait in silence.

6.10.2 Golden numbers

`--verify` checks the section 9 baselines and, once the engines have run, these figures. They are MAIIC's own, so a bootstrap on any server either reproduces them or fails:

Golden number	Expected	Source
ECL at 31 December 2025, total and by stage	The audited impairment allowance in the 2025 financial statements	The AFS mapping workbook (`TB_AFS_MAP`), the impairment lines; the signed statements
ECL at 31 December 2024	The audited 2024 allowance	The same workbook's December 2024 column
Contractual interest 2025, by loan GL	MWK 5,293,988,207.06 for January 2025 to July 2026 (section 9), split by month	The ledger
Interest income 2025 against the trial balance	The year-to-date tie, 83 of 83 account-months	Section 9
Loan balances by GL at each year-end	The audited figures: 2024 and 2025 by GL	The AFS mapping workbook
The revenue shift 2024 and 2025	Recorded on the first run that Dr Thom approves, then held as the number later builds must reproduce	The engine, once approved

The last row is how a golden number is born: the first approved run writes it, and from then on `--verify` fails any build that changes it without a governed reason.

Anything the bootstrap approves (a load, a build, a generated schedule, the seeded scenario set) is stamped with the approver label "System Bootstrap (automated data-readiness, not a MAIIC approval)", so that no one can mistake it for a sign-off by MAIIC; the maker-checker approvals of the Governance Centre and the register are never given by the bootstrap.

What it is for. The first installation on MAIIC's server; every UAT and acceptance round, which starts from a bootstrap so that the result is reproducible; Deloitte's copy; and the developers' own daily state, since a bootstrap with `--build --run-engines --verify` is the end-to-end test of the data foundation and of every engine on it. The monthly packs of section 6.5 are loaded by the feed, not by the bootstrap; the bootstrap

loads what is committed.

6.11 The EIR as at any date

A requirement in its own right: **the system shows the EIR computation for any loan as at any date the user names**, not only at month-ends and not only for the latest run.

- **What "as at a date" means.** The engine takes every posting in the landing zone dated on or before the date, the rate in force on the date, the governance values in force on the date, the schedule version in force on the date, and the take-on basis of 6.9, and produces for that date: the EIR in force and when it was last re-solved, the amortised cost, the gross carrying amount, the interest recognised to the date in the period and the year, the cumulative difference between the EIR interest and the contractual interest posted, the remaining expected cash flows, and the modification history. Each figure is shown with the inputs that produced it.
- **Any date, not only a period end.** A date inside a month uses the actual-day convention of section 3 (prior month-end balance, rate, days over 365) for the part-month. A date in a locked period reproduces the locked figures exactly, because a locked period keeps the settings and the schedule it was locked under; a date after the last loaded pack is refused with the last loaded date named, never estimated.
- **For the whole book.** The same view at portfolio level: the EIR interest, the contractual interest and the difference for the year to the date, by product, by GL and in total, which is the revenue shift Dr Thom asked for, at any date.
- **Where.** The Contract Profile carries a date picker ("View as at") beside the schedule; the Report Hub carries "EIR as at a date" for the book, with the Excel and PDF downloads; both read the same service (`EirAsAtService`), so a figure on the screen and in the download are the same computation.
- **The dates that matter now.** 31 December 2024, 31 December 2025 and 31 December 2026 are the year-ends Deloitte will ask for. Section 9 gains them as baselines once the first two are run.

7. The importers, reworked

Version 3 section 5 described five input files. The five remain, but four of them are now read from E-Banker's own tables. This section says what each importer does and the rules it enforces. Every importer refuses an ambiguous date and names the row (D19), logs what it ignored, and is idempotent.

7.1 The ledger importer (replaces Extract B and Extract C)

Lands `P1_01` in `ebanker_ledger` (section 6.3) and reads it from there: every posting on the in-scope accounts with its signed amount, transaction type, value date and narration. Rules: debits negative as received, never ABS'd; a reversal (306, 343, 901) is kept as its own row and paired to the posting it reverses by amount and date; the year-end type 120 is interest of December; a deleted row (nil value, deleted flag) is loaded and marked, not dropped. The ledger's running balance per account is the carrying amount for any date, and the reconciliation reads interest (303 and 120) from it. The old `GlInterestImportService`, which skipped annual rows, is retired.

7.2 The rate-history importer (new)

Reads three things and keeps them apart: the PLR master (`P2_06`) as the reference-rate series with its Reserve Bank effective dates; the monthly loan-book rate as the rate charged; and the rate set-up table (`P1_04`) as the register of when each account's rate was keyed and under which Interest Policy. Rules: the rate type is P floating, S and M fixed, read from `INTEREST_POLICY`; the Floating Flag is recorded verbatim and a disagreement is noted, never used to derive the type (the old `RATE_BASIS` mapping, which was inverted, is removed); the spread is charged rate less PLR, derived monthly, and a change in it is an event for the reset-or-modification rule (O17). The PLR master has one row (id 2) whose effective date was blank in the second export but present in the first; the first export is used.

7.3 The contract-master importer (reworked)

Reads `P1_02` and `P1_03` and loads the verbatim E-Banker code columns that P2 introduced (Interest Policy, Floating Flag, calculation base, instalment basis, moratorium type, the EMI type, the sanction-versus-balance basis). `MARGIN_PERCENT` is zero on every loan master row and is not used. The sanction amount and date are loaded; an account whose drawn balance exceeds its sanction is flagged for Credit (six today, all take-on balances; two need the offer letter).

7.4 The fee importer (two routes)

Section 7.8 sets out the whole path of a fee, from E-Banker to the EIR, with the sources, the delivery routes, the link to the loan, the checks and the settings. This section keeps the importer's own detail.

Loans disbursed through E-Banker: `P2_07`, the charges table, with the fee name, amount, posting date and income GL; the standing fee rulebook (25 rules) classifies each as integral or not. Take-on loans: Tamanda's template. Both land in the same table with their source recorded. The control total for both routes is the fee income GLs, 4871 and 4873, not the account ledger: about 44 percent of 2025 fee income, and nearly all of it from March 2026, was credited to income in the drawdown batch with no fee leg on the loan account (section 3.7). The importer therefore reconciles each month's fees to the GL spools and flags a GL fee line it cannot place on a loan.

Built 11 October 2026: the fee GL check. After every fee import, `FeeGlReconciliationService` takes each year the file covers and sets every fee held for that year (all imports, not only this file) beside the trial balance movement of its GL, by month and for the year, inside the `recon_tolerance` band. The GL of a line is its `gl_account_ref` when that names a fee GL, otherwise 4873 for an arrangement fee and 4871 for a legal fee; a PAID line (a cost MAIIC paid out, 6752) is left out of the income GLs and listed apart; a line with no date is reported because it cannot be placed in a month. The monthly GL figure is `TrialBalanceMovementService`'s, so December 2025 is read pre-closing from the AFS workbook (the 2025 year then totals 171,128,135.00 and 311,342,792.00, as in section 3.7). The result shows on the import (EIR Data Intake) and on Fee Classification, which gains an **Import fees** button. It reports and never blocks, and the Governance Centre setting `fee_gl_reconciliation` switches it off. A new permission, `eir.fees.import`, opens EIR Data Intake for the fee type only and Fee Classification read-only, so Finance can load fees by loan account without the broad settings permission; classifying and reviewing still need settings. The fee sample file now carries 4873, 4871 and 6752 in `gl_account_ref`, with a received legal fee and a paid legal cost. Tests: `tests/Feature/Eir/FeeGlReconciliationServiceTest.php`. The `COLLECTION_TYPES` list in `EirRevenueService` wrongly includes 'Interest' and treats an interest charge as cash collected; it is corrected.

7.5 Expected cash flows (generated, not imported)

The chart exists for 75 of 184 accounts and is unreliable where it exists, so the version 1 schedule is generated from the loan's terms on actual dates (P4) and compared with the chart and with the LOS schedule as references, as D20 and O7 provide. Drawdowns come from `P3_12`, with the tranches.

7.6 The take-on loader (new)

As section 6.9: the workbook landed as `takeon_blocks` and `takeon_schedule_lines`, the gates, the build of `contract_takeon` under the `takeon_history_basis` setting, and the Take-on Schedules screen.

7.7 The trial balances: landed with the rest, and the GL opening balances

The trial balance is the check on everything else, so it enters by the same door as everything else. The 20 monthly files, the December 2025 AFS bridge (the 10 September 2026 version is the one loaded; the 19 August version held its amounts as text) and the mapping of every GL line to the audited accounts are committed under `docs/bootstrap/trial-balances/` and landed as `ebanker_trial_balances` (section 6.3); each later month arrives as one more file in the monthly pack, with its hash in the manifest. `gl_trial_balance_lines`, which the reconciliation reads, is derived from the landed files by the build, and `gl_account_scope` still says what each GL code is. The two rules of the September importer are unchanged: a P&L balance is cumulative year-to-date and is turned into a month on read (January taken whole); a balance-sheet balance is never differenced. `GL_02` is landed as the small table of keyed GL opening balances so that the bridge shows the two FInES differences as what they are. The ties the trial balance gives (section 9: the loan GLs to the cent, the FInES differences, interest income year-to-date on 83 of 83) are run by `eir:bootstrap --verify` like every other baseline.

The two charts of accounts (added 11 October 2026). E-Banker's trial balance is not the ledger the accounts are prepared from. Finance keeps the statutory books in QuickBooks, and the AFS mapping workbook's "TB December 2025" sheet maps every E-Banker line to a QuickBooks code. For the lines the engine touches:

QuickBooks (accounts)	E-Banker (trial balance)	2025, MWK	What changes between them
4206 Interest on Term Loans-MAIIC	4215 + 4216 + 42019	2,106,561,165.01	The audit added the interest-in-suspense reversal (21,865,867.35); the accounts then presented 259,317,000 of arrangement fees here, a manual adjustment recommended by the auditors from an extrapolated sample and not posted, because E-Banker has no EIR calculation (section 3.7)
4209 Term loan interest-FInES	4218 + 4219	447,829,531.72	Nothing
4870 Services Income: 4871 Legal fees	4871	171,128,135.00	Nothing in 2025
4870 Services Income: 4873 Arrangement fees	4873	311,342,792.00	Shown as 52,025,792 in the accounts after the presentation above
1320 Interest Suspense	1320	21,865,867.35 (credit)	Reversed to 4206 by the audit

The reconciliation therefore runs in two steps: E-Banker's ledger to E-Banker's trial balance (the ties of section 9), then E-Banker's trial balance to the accounts through this mapping, with each audit adjustment as its own named line. The mapping is seeded in `gl_account_scope` (chart `QUICKBOOKS`, codes 4206, 4209 and 4870) and the full 191-row map is landed with the AFS bridge. E-Banker's own year-end posting, the re-rating batch of 31 December (section 3.4), is expected every year and sits inside the E-Banker figures, not among the audit adjustments.

7.8 Fees: how they are ingested and reconciled (added 11 October 2026)

In plain language. A fee is part of a loan's effective interest rate, so every arrangement and legal fee has to reach the engine once, on the right loan, at the right amount and date. MAIIC holds the same fee in several places: the GL fee accounts (4871, 4873), the loan account in E-Banker, the loan origination system's charges table, Finance's own workbook, and the offer letter. Only the GL posting is the accounting record; the others are evidence of it. This section says which record the engine takes, how it arrives, how it is tied to its loan, how it is proved against the trial balance, and which choices MAIIC can change. The findings behind it are in section 3.7 and in the note `docs/MAIIC_Fees_2025_Workbook_GL_and_Design_2026-10-11.md`.

7.8.1 How E-Banker records a fee

The fees are read off the offer letter, keyed by Finance and deducted from the amount advanced: a facility of 100 million with 5 million of fees pays out 95 million and is carried at 100 million. E-Banker posts it as one balanced batch on the drawdown date: the loan account is debited with the gross amount, the bank is credited with the cash paid out, and 4871 and 4873 are credited with the fees. The fee therefore appears in two places (on the loan and in income) because a double entry has two sides; it is not recorded twice. The loan side is written in one of two ways: a separate fee debit on the loan account, type 301 with a fee narration (about 56 percent of 2025 fee income), or one gross drawdown line with the fee inside it (the rest, and nearly all of 2026). The GL printout of 4871 and 4873 carries no account number; the link to the loan is the batch, which every leg of the drawdown shares.

7.8.2 The sources, and which one counts

Source	What it holds	How it arrives	Role
GL fee postings by loan (`GL_04a`, `GL_04b`)	Every posting to 4871, 4873, 4874 and 4875, and every other leg of the same batches	The E-Banker feed (7.8.3)	**The source of record** once landed: it is the GL, so it ties to the trial balance by construction, and it carries the posting date
Finance's fee workbook	The fees by loan account for a year (for 2025: `MAIIC EIR Assessment as of 31st December 2025 with account numbers.xlsx`, 34 loans, ties to the GL within 800)	EIR Data Intake, type fees, by Finance (`eir.fees.import`)	The source of record for 2025 until `GL_04` is landed; afterwards a cross-check
LOS charges (`P2_07`, LOS_DISB_CHARGES_POST)	What the origination system captured at application, with no date	The E-Banker feed (the whole table is RUN 5 of pack 4)	Evidence only: it differed from what was posted on 9 of 21 loans in 2025
Extract A fee columns	arrangement_fee and legal_fees on the contract master	EIR Data Intake, contract master	Evidence only
Take-on fees	Fees of the 109 loans taken on in July 2024, which E-Banker never posted	The take-on workbook or the Take-on Schedules screen (section 6.9)	The only record for those loans; they never compete with the others
Offer letters	The contract	Read by Credit and Dupleix	Evidence (decision D20)

One fee, one source. The engine reads a contract's fees from the highest source that holds them, in the order of the governed setting `fee_source_precedence` (7.8.6). A line from a lower source is kept and shown, marked `SUPERSEDED`, and never enters the EIR; where its amount differs from the source of record, Fee Classification lists the difference as an exception for Finance. This is what stops the same fee being counted twice: today the bootstrap loads `P2_07` into `contract_fees`, and importing Finance's workbook as well would count the fees of 21 loans twice and overstate the EIR.

7.8.3 How the GL fee postings arrive

`GL_04a` and `GL_04b` are two read-only queries in the register (section 6.4). They arrive like every other extract, by the route in force (`ebanker_feed_route`, section 6.5):

Route	What happens	Status
1. Manual pack (in force)	Barry runs the queries in SQL Developer (RUN 6 of pack 4), exports two CSVs and sends them; they are uploaded on Data Foundation, E-Banker Feed	Ready. The first run covers 1 January 2024 to date
2. Scheduled export	A Windows scheduled task at MAIIC runs the export script on the first working day after month-end and writes the pack to a shared folder that the system polls (`eir:poll-feed-folder`)	The script template already carries `GL_04a` and `GL_04b`; needs a read-only Oracle account and the folder from ICT

Route	What happens	Status
3. Push by API	The same script posts the zipped pack to the system with a token	Built; needs the token issued
4. Direct read over the VPN	The system reads E-Banker with a read-only account	Needs Dr Thom and ICT
5. Vendor view or API	The vendor exposes the same rows	Not started

After the first run the queries are incremental: postings with a `CUMVOUCH_DET_ID` above the watermark, plus the two months before the period again, so that a back-dated correction is caught. From the October 2026 month-end they are part of every monthly pack with `M09_01` to `M09_03`. Landing is unchanged: the pack passes the gates and the rows are kept verbatim in the landing zone, keyed on `CUMVOUCH_DET_ID`.

7.8.4 Tying each fee to its loan

The build joins `GL_04a` to `GL_04b` on the batch (transaction date, batch type, batch number) and writes one fee line per loan, fee type and posting, with the GL code, the amount, the posting date, the batch and the voucher, into `contract_fees` with the source `E-Banker GL fee posting`.

Case	Rule
One loan account in the batch (most drawdowns)	The fee belongs to that loan, whether the loan shows it as its own line or inside the gross drawdown
Several loans in the batch (2 January 2025: Ebenezer Midian and Doorbell Services)	Split by matching each loan's own fee debit; where the loan has none, by the narration; what cannot be split is held as an exception, never guessed
No loan account in the batch (a fee paid in cash, or a loan outside the eighteen schemes, such as the BRFF fees of June 2026)	Held as an unassigned fee line and shown on the reconciliation; never dropped
A correction or reversal (Mchinji: 3,020,000 reversed on 24 October 2025 and 2,920,000 posted)	Netted on the loan, both lines kept for audit

A fee debit is not a drawdown. The disbursement builder (`ContractInputsBuildService::disbursements`) today turns every type-301 debit on a loan account into a tranche. A type-301 line in a batch that also credits a fee GL, or whose narration names a fee, is instead the fee capitalised into the loan: it is added to the amount drawn at origination and never becomes a tranche of its own; a reversal reduces it. With that, the initial net investment is the gross amount drawn less the fees, which is the cash MAIIC paid out.

7.8.5 How the fees are reconciled

Check	When	What it proves	Status
Fee GL check (`fee_gl_reconciliation`)	After every fee import, and always on Fee Classification	The fees held for each year against the trial balance movement of 4871 and 4873, by month and for the year, inside `recon_tolerance`; PAID lines and undated lines listed apart; reports, never blocks	**Built 11 October 2026** (7.4)
Fee GL gate on the monthly pack	When a pack lands	The month's movement on each fee GL in `GL_04a` equals the trial balance's	To build (P4b)
Assignment completeness	After the batch link	Every `GL_04a` line is on a loan, or listed as unassigned or held, with its reason	To build (P4b)
Source cross-check	After each load	Where a lower source differs from the source of record: the loan, the fee type, both amounts and the difference, on Fee Classification	To build (P4b)
Two-chart bridge	Year end	The fees in E-Banker's trial balance to the audited accounts (section 7.7)	Mapping seeded; bridge to build

For 2025 the arithmetic is already known: Finance's workbook ties to the GL at 311,342,792 (arrangement) and at 171,128,135 less 800 (legal: one keying slip on Chikwendeni General Dealers), and every GL fee line of 2025 traces to a loan in it. The workbook's own check rows show a doubled figure because row 46 sums a range that includes the row-39 subtotal; that is a spreadsheet range, not a second posting.

7.8.6 The choices MAIIC can make

Setting (Governance Centre)	Options	Default	Status
`fee_gl_reconciliation`: fee imports checked against the fee GLs	On; Off	On	Built
`fee_source_precedence`: which source a contract's fees come from	GL posting, then Finance workbook, then LOS capture, then Extract A; or Finance workbook first; or LOS capture first	GL posting first	To build (P4b)
`ebanker_feed_route`: how the packs, `GL_04` included, arrive	Routes 1 to 5 (7.8.3)	Route 1	Built
`recon_tolerance`: the band for every reconciliation	As in section 4.2	1 percent, floor MWK 1	Built
`takeon_history_basis`: how the take-on fees are used	As in section 6.9	Recompute where block and fees exist	Built

7.8.7 What Finance does, in order

1. For 2025, import the fee workbook through EIR Data Intake (type fees) with the GL posting date as `transaction_date` and 4871 or 4873 in `gl_account_ref`. Dupleix prepares the mapped file from the workbook and the GL lines, with the source `MAIIC Finance EIR assessment 2025`. This waits for `fee_source_precedence`, so that it supersedes the `P2_07` lines rather than adding to them.
2. Check the fee GL card on Fee Classification: every month of 2025 agrees, or the difference is named.
3. From the October 2026 month-end nothing is typed: the fees arrive with the monthly pack through `GL_04`, are tied to their loans by the batch, and are proved against the trial balance by the gate and the check.
4. Classify each line (integral or not) and have it reviewed, as today; only reviewed integral lines enter the EIR.
5. For the take-on loans, fill the fee columns of the take-on workbook or of the Take-on Schedules screen (section 6.9): one of the two, not both.

8. The build plan from here

Phases P1 to P4 are built. The order from today, with what each one waits for:

Phase	Content	Waits for
P4b Ingestion and importer rework (sections 6 and 7)	The landing zone, the pack contract and gates, route 1 and the feed screen, the build by all three methods (the report importer landed and gated); the take-on schedules landed, gated and built (6.9); `EirAsAtService` (6.11); `eir:bootstrap` with the committed inputs and `--verify` (6.10); the rate-history, contract-master, fee and take-on importers; the four code corrections; route 2's script once the read-only account exists; added 11 October 2026: `GL_04` in the monthly pack with the fee-GL gate (6.4), the fee importer's reconciliation to the fee GLs (7.4; built 11 October 2026 with its Governance Centre switch and the `eir.fees.import` permission), and the two-chart reconciliation to the accounts (7.7); the fee path of section 7.8: `fee_source_precedence` with superseding, the batch link of `GL_04` to the loan, fee debits capitalised rather than made tranches, the assignment and cross-source checks; added 12 October 2026: the write-off rule of section 3.8 in the loan book build, the LGD engine, the EIR and the RBM return, with status H mapped to written off; the reschedule register built from `RS_02` (section 3.9)	Nothing; route 4 waits for Dr Thom and ICT
P5 Floating resets	Reset detector from the PLR series writing `rate_reset_events`; a spread change as a separate event; maker-checker intake; a reset inside a locked period refused; prospective re-estimation under B5.4.5	Nothing to build; O17 confirmed by Deloitte before the first reset is booked
P6 Arrears	Cash receipts from the ledger (exact); re-estimation under B5.4.6; IRR on actual expected flows	P5
P7 Restructuring	Version N+1 import; `contract_modifications`; the 10 percent test; lineage	The restructure register from MAIIC
P8 Month-end run and screens	Pipeline, period lock, Contract Profile with the as-at date picker, Rate Resets, Restructures, Drawdowns, Month-end Run, the EIR-as-at-a-date report; help articles; the journal proposal reading the true-up account (O10)	P5 to P7; O10
P9 Acceptance and UAT	T1 to T8 on MAIIC's data; the revenue shift per year, 2024, 2025 and 2026 to date, which is the output Dr Thom asked for by name; UAT with Finance	The fee template; D30 confirmed; the sign-offs
P10 Deployment and training	Install in MAIIC's environment; training; manuals; source code	P9

P4b and P5 start now. Fees and the Mega Farms decision are the two items that gate the number; everything else is engineering.

Status at 13 October 2026. P4b is built, with `EirAsAtService` (the EIR as at any date, by contract, product and GL, with the credit-balance flag and the months a loan has no revenue row). The month-end run of P8 is built as the Maintenance month-end job (D37) and the corrected roll-forward of section 3.10 is in it. The

re-estimation of floating-rate loans when the charged rate moves (P5, decision D38) and the arrears at the opening (part of P6, decision D39) are built: amortised cost ties to the ledger within 5.8 million at August 2026 (section 3.10). Still to build in P5: the reset events written to `rate_reset_events` from the PLR series with maker-checker intake and the refusal of a reset inside a locked period. Still to build in P6: re-estimation of expected cash flows under B5.4.6 when a loan is credit-impaired.

9. Acceptance baselines

The ties achieved this week become regression tests, run by `eir:bootstrap --verify` (section 6.10) and shown on the Baselines sheet of every audit workbook (section 12). A build that cannot reproduce them has broken something.

Test	Expected
Ledger running balance against the balance history, every month-end	2,533 of 2,549 account-months agree; the 16 that do not are Zaithwa Farms from May 2025
Ledger running balance against the stored loan book's carrying amount	2,264 of 2,264 agree
Ledger interest (303 and 120) against Extract C run 41	All 1,564 account-months; total MWK 5,293,988,207.06
Loan GLs against the trial balance, every month	1050101, 1050102, 1050401 to the cent; 1050201 +400,000.00; 1050202 -1,000,000.00
Interest income against the trial balance, year-to-date	83 of 83 comparable account-months
Contractual interest rebuilt from the previous month-end balance, rate and days over 365	1,595 of 2,047 within 0.1 percent; 1,828 within 1 percent
The month of a rate change	Whole month at the new rate on 217 of 229
Take-on postings of 31 July 2024	77 postings, MWK 8,297,388,309.25
Fee income GLs against the trial balance	4871 and 4873 year-to-date equal the trial balance in every month, January to November 2025 and January to July 2026; 2025 before the closing voucher 171,128,135.00 and 311,342,792.00
4206 against the audited accounts, 2025	E-Banker 4215 + 4216 + 42019 = 2,106,561,165.01; plus the suspense reversal 21,865,867.35 = 2,128,427,032.36 (final AFS TB); plus the fee presentation 259,317,000.00 = 2,387,744,032.36 (section 3.7)
Write-offs on GL 1064	22,286,434.32 in August 2024 (type 300, two accounts); 138,135,680.80 in January 2026 (memorandum, against 1876); 38,492,032.54 on 29 June 2026 (TRF 3, 20 accounts with status H); none of them read as a collection (section 3.8)
Year-end adjustments of 31 December 2025	28 postings: 22 debits 96,390,096.16; 6 credits 85,166,683.31; GL legs 4215 +32,956,675.55 and 4216 -21,733,262.70; December income movement on both GLs equal to the ledger to the cent
The monthly feed (September 2026, the first pack)	August re-pull identical to the first load; September balance change equals the ledger on 105 of 105 accounts
Collateral register	118 distinct securities, MWK 47.1 billion, after de-duplication on the security id; 58 of 89 active accounts covered
Take-on mapping	105 of 109 facilities; 98 of 100 schedule blocks linked; principal equals the take-on posting on 77
Loan book built by method B against the same month built by method A, every month-end from December 2024	2,264 of 2,264 carrying amounts agree; every difference is a flagged row with a named cause

10. Risks and how they are held

Risk	Effect	Held by
An extract is re-saved through Excel on its way to us	Dates scrambled, as in September	ISO-only importers that refuse and name the row; the session settings in every request; the files never opened in Excel at our end
The loan book is re-run after we load a month	A back-dated posting changes a carrying amount	The bootstrap keeps the latest run and prints every difference before overwriting; a locked period is never restated
The FInES GL openings are not corrected	A 600,000 net line in every reconciliation	Carried as a named line until Finance adjusts
The fee template is late	The 2024 and 2025 shift cannot be computed for the take-on book	Post-migration loans proceed on `P2_07`; the take-on figure is marked incomplete rather than estimated
A pack arrives by a new route before its gates are proven	A month derived from a file that did not tie	The gates are route-independent and run on every pack; a route is accepted only after one pack from it has passed on a copy
The daily accrual exists for 14 accounts only	The daily test cannot be run on the rest	The narration test covers all 2,047 postings
A governed default is confirmed late	Nothing stops; the month runs on the seeded value	The Governance Centre records which value each month ran under; a later change applies forward only

11. The user interface: the Dupleix-suite layout

11.1 In plain language

Every system Dupleix now delivers uses the same shape of screen, the Dupleix suite layout, so that a finance officer who has learned one of them can find their way around the next. MAIIC's system was built earlier, and its menu still follows the headings of the contract schedule ("Customer & Loan Data", "IFRS 9 Model Setup", "ECL Processing") rather than the way the work is actually done. Decision D24 adopts the suite layout for MAIIC.

The shape is simple. On the left is a dark navigation panel with six working groups, each with its own colour: **Data Foundation** (what comes in), **Governance Centre** (the rules and settings that govern the calculations), **Financial Modelling** (the engines), **Risk & Regulatory** (the regulatory views), **Monitoring** (the watch-lists and alerts) and the **Report Hub** (what goes out), followed by System Documentation and Administration. Across the top is a bar with the menu toggle, the financial period the system is working in, an appearance switch (light, dark, or follow the device), notifications and the user's menu. Every page opens with a header that says where you are (the group, then the page) and what the page is for. The panel can be folded to a narrow rail of icons when the screen is small or the user wants room for a wide table. Nothing about the calculations changes; the screens move to where a user would look for them.

11.2 What the layout is made of

Element	What it does	Source
Navigation panel (sidebar)	The brand at the top, then the two top-level links (Dashboard, Workspace), then the groups. One group is open at a time; the group that contains the current page opens by itself. Each group has an icon tile in its own colour, and the page in use is marked with a bar in that colour.	The suite's sidebar; MAIIC keeps its own recursive renderer so that a group can hold sub-groups (the modelling group needs them)
Icon rail	The hamburger in the top bar folds the panel to 68 pixels of icons; the choice is remembered in the browser. On a phone the same button opens the panel as a drawer over the page.	The suite's shell
Top bar	Menu toggle; a chip showing the financial period the system is working in and whether it is open or closed; the appearance switch (11.5); the notification bell; the user menu (profile, API tokens, log out). The decorative search box that does nothing today is removed.	The suite's top bar
Page header	An icon tile, the breadcrumb (group, then page) derived from the navigation tree and the current route, the page title, a one-line description, and a slot on the right for the page's actions. A page that already supplies its own header keeps it, inside this frame.	The suite's shell
Processing pill	The floating "Processing" and "Done" indicator while a request is in flight. Already in MAIIC; unchanged.	Already in MAIIC
Fail-loud error dialogue	The plain-language explanation of a 403, 419, 404 or server error. Already in MAIIC; unchanged.	MAIIC
Per-page help	MAIIC's help centre and the Help button on every page stay as they are. The guide text that names menu groups is re-worded to the new groups.	MAIIC

Everything in the suite's shell is adopted, including light and dark mode (11.5). MAIIC's pages were not written with a dark variant, so the dark mode is delivered in two steps: the shell and the shared styles first, then a sweep of the pages (11.9).

11.3 Where every screen lives

The tree is the single source of truth: the server holds it in `config/menu.php`, filters it by the user's permissions, and sends it to the browser, which renders the panel and derives the breadcrumb from it. Nothing is duplicated in the Vue code. Routes do not change; only the grouping, the group names and the order do. "Today" is the group each screen sits in now.

Group (colour)	Screen	Route	Permission	Today
Top level	Dashboard	`dashboard`		same
Top level	Workspace	`workspace.index`		same
Data Foundation (teal)	Clients	`clients.index`		Customer & Loan Data
	Loan Book	`loan_applications.loan-book`		Customer & Loan Data
	Imports	`imports.index`		Customer & Loan Data

Group (colour)	Screen	Route	Permission	Today
	E-Banker Feed (queries, loads, watermarks, build)	`eir-feed.index` (new, section 6.8)	eir.view; derive needs eir.govern	new
	Take-on Schedules (workbook, mapping, fees, build)	`eir-takeon.index` (new, section 6.9)	eir.view; build needs eir.govern	new
	Loan Portfolios	`portfolios.index`		Portfolio Setup
	Product Groups	`groups.index`		Portfolio Setup
	Sector Types	`industry_types.index`		Portfolio Setup
	Collateral Register, Types, Allocation	`collateral.register.index`, `collateral.types.index`, `collateral.allocations.index`		Collateral Management
	EIR Data	`eir-data.index`	eir.view	EIR & Revenue Recognition
	Drawdowns	`eir-drawdowns.index`	eir.view	EIR & Revenue Recognition
	Reference Rates	`eir-reference-rates.index`	eir.view	EIR & Revenue Recognition
	Macro Statistics (variables, data, scenario assumptions, import from the World Bank, the IMF and the RBM)	`macro-statistics.index` (rebuilt, section 13)	macro.view; import needs macro.manage	IFRS 9 Model Setup (Macro Elements)
Governance Centre (amber)	Governance Centre (the settings of section 4.2)	`eir-governance.index`	eir.govern	EIR & Revenue Recognition
	Accounting Rules	`eir-accounting-rules.index`	settings	EIR & Revenue Recognition
	Fee Classification	`eir-fee-classification.index`	settings	EIR & Revenue Recognition
	Staging & SICR Rules (sub-group): Quantitative Thresholds, SICR Groups Setup, SICR Alert Items	`stageing-rules.index`, `sicr-groups.index`, `sicr-items.index`		IFRS 9 Model Setup
	Scenario Sets (propose, approve, lock, versions, back-test, sensitivity; section 15)	`scenario-sets.index` (replaces `scenarios.profiles`)		IFRS 9 Model Setup
	Financial Periods	`accounting.financial_periods.index`		Administration
	Audit Trail	`audit-trail.index`		Administration
Financial Modelling (sky)	PD Model (sub-group): Transition Profiles, Monthly Probability, Cumulative Probability, Internal Grades	`transition-profiles.index`, `transition-matrices.index`, `transition-matrix-cummulative.index`, `internal-grading.profiles`		IFRS 9 Model Setup
	LGD Model (sub-group): Monthly LGD, Cumulative LGD	`loss-given-default.index`, `lgd-cummulative.index`		IFRS 9 Model Setup

Group (colour)	Screen	Route	Permission	Today
	Forward-Looking Model (sub-group): Correlation Finder (new, 14.4), Regression Analysis (repaired, 14.5), FLI Adjustments (new, 14.6), Weighted Forecast, Credit Loss Data, Adjusted Forecast	`fli-correlation.index`, `regression.index`, `fli-adjustments.index`, `macro-forecast-weighted.index`, `credit-loss-data.index`, `forecasting.manual`		IFRS 9 Model Setup (Macro Elements moves to Data Foundation as Macro Statistics; Scenario Profiles becomes Scenario Sets under Governance)
	Management Overlays (sub-group): Economic Scenarios, External Calculations, Calculation History	`fli.scenarios.index`, `fli.external.index`, `fli.external.list`		IFRS 9 Model Setup
	ECL Calculation	`expected-credit-loss.index`		ECL Processing
	EIR Calculations	`eir-calculations.index`	settings	EIR & Revenue Recognition
	Coverage & Blockers	`eir-coverage.index`	eir.view	EIR & Revenue Recognition
Risk & Regulatory (indigo)	Stress Testing	`stress-testing.index`		Reports
	Sensitivity	`ifrs9-reports.sensitivity`		a tile in the hub
	IFRS 9 Disclosure	`ifrs9-reports.fs-disclosure`		a tile in the hub
	RBM Classification	`ifrs9-reports.rbm-classification`		a tile in the hub
	IFRS 9 vs RBM	`ifrs9-reports.ifrs9-vs-rbm`		a tile in the hub
	Concentration	`ifrs9-reports.concentration`		a tile in the hub
Monitoring (emerald)	SICR Trigger Alerts	`sicr-triggers.index`		IFRS 9 Model Setup
	SICR Trigger Report	`ifrs9-reports.sicr-trigger`		a tile in the hub
	Early Warning System	`ifrs9-reports.ews`		a tile in the hub
	Data Quality	`ifrs9-reports.data-quality`		a tile in the hub
Report Hub (violet)	IFRS 9 Reports (the full catalogue of 30)	`ifrs9-reports.index`		Reports
	EIR as at a date (the book, with downloads)	`eir-as-at.index` (new, section 6.11)	eir.view	new
	Executive Summary	`ifrs9-reports.executive`		a tile in the hub
	AI Commentary	`ifrs9-reports.ai-narrative`		a tile in the hub
	ECL Reconciliation	`reports.ecl-reconciliation`		Reports
	GL Reconciliation (EIR)	`eir-reconciliation.index`	eir.view	EIR & Revenue Recognition
	Loan Book Reconciliation	`reports.loan-book-reconciliation`		Reports

Group (colour)	Screen	Route	Permission	Today
	Disbursements (Vintage)	`reports.disbursement-report`		Reports
System Documentation (slate)	User Manual, Administrator Manual, Technical Manual, Installation Guide	`help.index`, `help.admin`, `docs.technical`, `docs.installation`		same
Administration (rose)	User Management, Roles & Permissions, Support Tickets, Settings	`users.index`, `users.roles.index`, `tickets.index`, `settings.index`		same, less the two that moved to Governance

Three rules behind the placement. A screen that **captures or shows what came in** is Data Foundation, whichever module uses it, so the EIR's data, drawdowns and reference rates sit beside the loan book and the collateral. A screen that **sets a rule the engines obey** is Governance Centre: the EIR settings, the accounting and fee rules, the staging thresholds, the periods and the audit trail that proves who changed what. A screen that **produces a figure** is Financial Modelling; one that **re-presents figures for a regulator or a reader** is Risk & Regulatory or the Report Hub. The hub's catalogue of 30 reports is unchanged; the eleven listed above are also reachable from the panel because they are the ones used weekly.

11.4 Behaviour, stated precisely

- **Permissions.** The server drops any entry whose permission the user lacks, and any group left empty, before the tree is sent (`HandleInertiaRequests::visibleMenu`, unchanged). A user never sees a link that would answer 403.
- **Active state.** An entry is active when the current route name equals its route (or its `route_check`). The group containing the active entry is open on page load and its header carries the group colour; one group is open at a time; the user may open another, which closes the first.
- **Breadcrumb.** Derived on the client from the tree and `route_name`: top-level entries give one crumb; a grouped entry gives group, then page; a sub-grouped entry gives group, sub-group, then page. A page outside the tree (an edit form reached from a list, for example) shows the crumb of the nearest list by its `route_check`.
- **Collapse.** Stored in the browser as `maiic.sidebar.collapsed`; on a screen narrower than 768 pixels the toggle opens the drawer instead. In the rail, each group shows its icon tile and, under it, the icons of its leaves with the label as a tooltip; sub-groups are flattened into their parent's icons.
- **Period chip.** The top bar shows the latest financial period and "Open" or "Closed" from its `closed` flag, shared from the server as `currentPeriod`. It is a display; it does not change the period.
- **Page header.** `title` and `description` may be passed as props; otherwise the title is the active entry's name and the description is empty. The `#header` slot, where a page provides one, renders inside the header frame in place of the derived title; the `#actions` slot renders on the right.
- **Colours.** Group colours are Tailwind families (teal, amber, sky, indigo, emerald, violet, slate, rose) written in full in one accent map so that the build includes them. The brand remains MAIIC navy and gold on the dark gradient.

11.5 Light and dark mode

Every user chooses how the system looks, and the choice follows them.

- **Three settings.** Light, Dark, and Follow the device. The switch is in the top bar and cycles through the three; the same choice is on the user's profile page. The default for a new user is Follow the device.

- **How it is applied.** Tailwind runs in class mode (`darkMode: 'class'`): the whole system is dark when the `dark` class is on the `<html>` element and light when it is not. Nothing else decides the theme, so there is one switch to test.
- **No flash of the wrong theme.** A four-line script in `app.blade.php` runs before the page paints: it reads the saved choice from the browser (`maiic.theme`), falls back to the device setting when the choice is Follow the device or absent, and sets the class. The Vue code then mirrors that state in one shared composable (`useTheme`) so the top-bar switch, the profile page and any other consumer stay in step.
- **Where the choice is kept.** In the browser, so that the first paint is right, and on the user's record (`users.theme_preference`, values `light`, `dark`, `system`), so that it follows the user to another machine: on login the server's value is written to the browser; when the user changes it, the browser value is written and the server is told. If the browser's storage is unavailable, the theme still applies for the session.
- **What dark mode must look like.** The sidebar keeps its dark gradient in both modes (it is dark by design). The page background, cards, tables, inputs, buttons, badges, modals, charts and the help panels carry a dark variant: slate backgrounds, light text, the group colours unchanged, MAIIC gold kept for the active marks. Contrast meets WCAG AA in both modes; no screen may show light text on a light ground or dark on dark.
- **How the pages get there without rewriting 281 components.** The shared classes the pages already use (`.card`, `.th`, `.td`, `.primary-btn`, `.secondary-btn`, the form inputs, the badges, the flash messages) are given their dark variant once in `app.css`, so most pages inherit the theme. The pages that style elements directly are then swept one group at a time (11.9), each page checked in both modes before it is signed off. Printed and exported outputs (PDF, Excel) are always rendered in the light palette, whatever the screen shows.

11.6 What changes in the code, and what does not

Change	Where
The tree regrouped as in 11.3, with an `accent` key on every group	`config/menu.php`
Accent map and group colours in the renderer; icon-rail mode	`resources/js/Jetstream/DropdownMenu.vue`, `SidebarNav.vue`, a new `resources/js/navAccents.js`
Collapse state, drawer, page header with breadcrumb, period chip; the dummy search removed; the dead consultation-channel code removed	`resources/js/Layouts/AppLayout.vue`
`currentPeriod` and the user's `theme_preference` shared with every page	`app/Http/Middleware/HandleInertiaRequests.php`
Tailwind in class mode; the no-flash script; the `useTheme` composable; the top-bar switch; the profile setting and its endpoint; `users.theme_preference`	`tailwind.config.js`, `resources/views/app.blade.php`, `resources/js/composables/useTheme.js`, `AppLayout.vue`, the profile page, one migration and `ProfileController`
Dark variants of the shared classes, then of the pages that style directly, group by group	`resources/css/app.css`, then the page components under `resources/js/Pages`
Help-centre guides that name a group (fifteen passages) re-worded	`database/seeders/data/help_user_content`, `help_admin_content`
User and Administrator manuals: navigation chapter re-written and screenshots retaken	`docs/manuals`
A test that every leaf in the tree names a registered route and that every group has an accent	`tests/Feature/NavigationTest.php` (new); `SystemDocsTest` unchanged

Not changed: any route, controller, page component, permission or calculation. The three icons the groups need and MAIIC does not yet register (shield, bell, balance-scale) are added to the Font Awesome library.

11.7 Acceptance

1. Every entry in 11.3 is reachable from the panel by a user with the right permission, and absent for one without it.
2. The breadcrumb on each of the 60 screens reads group, then page, as 11.3 lists them.
3. The rail, the drawer and the open-one-group rule behave as 11.4 states, on a 1366-pixel laptop and a phone.
4. The help centre and the two manuals name no group that no longer exists.
5. `SystemDocsTest` and the new navigation test pass; the EIR suite is unaffected.
6. The appearance switch cycles Light, Dark and Follow the device; the choice survives a reload, a new tab and a login on another browser; there is no flash of the wrong theme on load.
7. Every screen in 11.3 is checked in both modes: no unreadable text, no white panel on a dark page, charts and modals themed; PDF and Excel outputs unchanged.

11.8 Order of work

UI-1 the tree and the accent map (half a day); UI-2 the shell: rail, header, period chip, the appearance switch, the no-flash script and the shared dark classes (one day); UI-3 help text and the manuals' navigation chapter with new screenshots in both modes (half a day); UI-4 the walk-through of 11.7 items 1 to 6 on a copy of the database; UI-5 the page sweep for dark mode in the order of 11.9 (two days), with item 7 signed off group by group. UI-1 to UI-4 are done before P8, so that the new EIR screens of P8 are placed in the suite layout, and in both modes, from the start; UI-5 may run beside P5 to P7. All of it follows P4b, which does not touch the interface.

11.9 The dark-mode sweep, in order

The pages are swept in the order a user meets them, and each group is signed off in both modes before the next starts: (1) Dashboard, Workspace and the login pages; (2) the Report Hub and the IFRS 9 reports, which are what the CFO and the auditors open; (3) Data Foundation; (4) Governance Centre, including the Governance Centre screen itself; (5) Financial Modelling; (6) Risk & Regulatory and Monitoring; (7) System Documentation and Administration. The EIR screens are written with both variants from the start and are not part of the sweep.

12. Compliance audit workbooks

12.1 In plain language

An auditor's first question is not "what is the number" but "show me where the standard says so, and show me the system doing it". Dupleix's compliance-audit engine, in production in the suite, answers that with one audit workbook per standard or directive: every section of the directive on its own row, what the system does about it, where to see it, a status, and a place for the reviewer to sign. The same engine builds the Excel, a Markdown twin and a PDF from one data file, so the three can never say different things. Decision D25 adopts it for MAIIC, with two additions the EIR work makes possible: each row names the Governance Centre setting that governs it, and each row names the test that proves it.

The result is five workbooks, a register in the system where MAIIC signs each row, and an auditor's pack per period that bundles them with the figures. Deloitte gets a document they can walk from paragraph to screen to test; Dr Thom gets a count of what is done, partly done, outstanding, not applicable or waiting on evidence, which is the project's status in one line.

12.2 The five workbooks

Workbook	What the rows are	Reviewer
IFRS 9: the EIR and amortised cost	The paragraphs the engine implements: the Appendix A definitions of effective interest rate, amortised cost and gross carrying amount; 5.4.1 to 5.4.4 (interest at the EIR; Stage 3 on the net amount, 5.4.1(b)); B5.4.1 to B5.4.7 (fees that are integral, the floating-rate reset, re-estimation of cash flows); 5.4.3 and B5.4.6 (modification gain or loss); 3.3.2 and B3.3.6 (derecognition and the 10 percent test)	Dr Thom, then Deloitte
IFRS 9: impairment	5.5 and B5.5: staging, significant increase in credit risk, 12-month and lifetime losses, forward-looking information, write-off; the ECL module already built	Dr Thom, then Deloitte
IFRS 7 and IAS 1: presentation and disclosure	IAS 1.82(a), interest revenue calculated using the EIR shown as its own line; IFRS 7.20 and 7.35A to 7.35N, the credit-risk disclosures; each row mapped to the disclosure report that produces it	Deloitte
RBM classification and provisioning	The Reserve Bank directive behind the "IFRS 9 vs RBM" report, section by section, with the report line that answers each	MAIIC Risk
Contract Schedule 1	Each deliverable and acceptance item of the implementation agreement, with the screen, document or test that discharges it	Dr Thom, at acceptance (P9)

12.3 What a workbook contains

Four sheets. The first three are the engine's standard shape; the fourth is added for MAIIC.

Sheet	Content
Contents	The standard's own order of sections, each hyperlinked to its row on the Audit sheet; the status counts at the top
Audit	One row per section of the standard, eleven columns (12.4), a status drop-down with colour coding, and conditional formatting that flags a "Done" with no test named
Findings	What needs a decision or a fix: number, reference, finding, what was found, impact, recommended action, owner, status
Baselines	The acceptance ties of section 9, one per row: the test, the expected figure, the system's current figure read at generation, and a live PASS or FAIL formula. The MAIIC counterpart of the engine's golden-numbers check

12.4 The eleven columns of the Audit sheet

The engine's nine standard columns, then two MAIIC additions.

#	Column	What goes in it
1	Reference	The paragraph or section number of the standard
2	Section name	Its heading
3	What it requires	The requirement in one or two plain sentences
4	Status	One of: Done; Partially done; Outstanding; Not applicable, documented; Evidence needed from MAIIC
5	What the engine does	The behaviour, and the evidence checked when the status was set
6	General comment	Anything a reader needs that the other columns do not carry
7	Compliance comment	How the behaviour satisfies the requirement, or why it does not yet
8	Where to see it	The screen (as a route the register turns into a link) and the report or export

#	Column	What goes in it
9	Reviewer sign-off	Initials and date; in the register this is captured under maker-checker, not typed
10	**Governance setting**	The key of the setting in section 4.2 that governs the behaviour (for example `rate_change_classification` on B5.4.5), so a reviewer sees which choice each paragraph turned on
11	**Test that proves it**	The PHPUnit test, by class and method, that fails if the behaviour changes; a row may be Done only if this column is filled

The five statuses are the engine's standard ones: Done is implemented and visible in an approved output; Partially done is mechanics in place with a parameter, input or piece of evidence still differing from the requirement; Outstanding is required and not built; Not applicable carries its reason and reference; Evidence needed means the system is ready and MAIIC must supply a document, minute or dataset.

12.5 Where it lives in the system

Place	What is there
Governance Centre, Compliance Audits	The register. One card per workbook with its status counts and the three downloads (Excel, PDF, Markdown). Opening a card shows the rows; a reviewer with the govern permission sets a row's status and signs it, a second person approves, and the audit log records both. Column 8 renders as a link to the screen. The workbook is generated from the register, so the signed state is the database, never a file someone edited
Report Hub, Auditor Pack	Per period: the Deloitte export (O12), the five workbooks as at that period, and the Baselines sheet, bundled in one archive with a checksum per file and a manifest, the way the suite archives a regulatory return. This is what is handed to Deloitte
Governance Centre, Audit Trail, Audit Trace	The suite's per-record trace: open any contract and see, oldest to newest, every change to its schedule, every reset and modification, and the governance values each of its months was run under. The workbook states the rule; the trace shows the rule applied to one loan

12.6 How it is built

- **The engine is ported, not rewritten.** The suite's `tools/compliance/` (`audit_workbook.py`, `build_audit.py`, `md_to_pdf.py`) becomes `tools/compliance/` in MAICC-IFRS9, with the two columns and the Baselines sheet added. Each standard is a data module in `tools/compliance/standards/` exposing `META`, `ROWS`, `FINDINGS` and `BASELINES`; the builder validates them (eleven fields per row, a known status, no duplicate reference, a reason on every Not applicable, a test on every Done) and refuses to write a misleading workbook.
- **One source, three outputs, one register.** The data modules are the source. The builder writes the Excel, the Markdown and the PDF to `docs/compliance/`; a seeder loads the same rows into `compliance_audit_rows` for the register; the register's generation step reads the signed state back and rebuilds the three files. Nothing is typed twice.
- **The Baselines sheet reads the system.** At generation the builder runs the section 9 checks against the database and writes the current figure beside the expected one; the PASS or FAIL formula lives in the sheet so Deloitte can see it recompute.
- **Tables and code.** `compliance_audits` (one per workbook: key, title, standard, reviewer), `compliance_audit_rows` (the eleven columns, status, signed_by, signed_at, approved_by, approved_at), `compliance_findings`; `ComplianceAuditService`, `ComplianceAuditController`, pages under `Pages/Governance/Compliance`; the trace page `Pages/Audit/Trace.vue` ported with the EIR subjects (contract, schedule, reset, modification, setting); the pack builder `eir:build-auditor-pack {period}`.
- **The IFRS 9 EIR module is drafted first**, from this specification: every paragraph it cites already maps to a service and a test, so its rows can be written now and signed as the phases land. The impairment, disclosure and RBM modules follow from the existing reports; Schedule 1 from the contract.

12.7 Acceptance

1. Each of the five workbooks builds from its module in the three formats, and the Markdown and Excel carry identical rows.
2. The builder refuses a Done row with no test, a Not applicable with no reason, and a duplicate reference.
3. In the register, a status change and a sign-off need two people, and both appear in the audit log with the old and new values.
4. Every column 8 link opens the screen it names for a user with the permission; every column 11 test exists and passes.
5. The Baselines sheet shows PASS on every section 9 tie against the production copy.
6. The auditor's pack for a period contains the export, the five workbooks and the manifest, and every checksum in the manifest verifies.

12.8 Order of work

CA-1 port the engine and add the two columns and the Baselines sheet (one day); CA-2 the IFRS 9 EIR module, drafted from this document (one day, then kept current as P5 to P7 land); CA-3 the register, the trace and the pack (two days); CA-4 the four remaining modules (two days, Schedule 1 last, at P9). CA-1 and CA-2 start with P4b; CA-3 sits with P8, since the register and the pack are screens of the suite layout; CA-4 is finished before UAT.

13. Macro statistics: ingestion from the World Bank and the IMF

13.1 In plain language

The forward-looking part of IFRS 9 (B5.5.49 to B5.5.54) asks MAIIC to adjust its expected credit losses for what is reasonably expected to happen to the economy: growth, inflation, the exchange rate, interest rates, the harvest. Today those figures are typed into the system by hand from whatever source the analyst had open, with no record of where a number came from or when. The Dupleix suite does this differently: the system fetches the published series itself from the World Bank's open data service and the IMF's World Economic Outlook, shows the analyst what it found, and writes it only when the analyst says so, with the source, the address and the time kept against every figure. Decision D27 adopts that for MAIIC.

13.2 What MAIIC has today

The module exists and its tables are sound. `macro_statistics` holds the definition of each series (code, name, unit, frequency, how many historical and forecast periods, source, website link, active flag); `macro_statistics_data` holds one value per series, period and scenario, with an actual-or-forecast flag, an FLI flag, confidence bounds, assumptions and a free-text source, unique on series, period and scenario. Six screens use them under IFRS 9 Model Setup: Macro Elements (manual entry), Scenario Profiles, Weighted Forecast, Credit Loss Data, Adjusted Forecast and Regression Analysis, and the Management Overlays group holds the economic scenarios and external calculations.

What is missing is everything before the first screen: no external source, no preview before a write, no record of where a value came from, no command a bootstrap or a scheduler could run, and no rule for what happens when two sources disagree. The scenario-on-the-row design is in one respect ahead of the suite's and is kept.

13.3 What the suite does, and is adopted

Element	What it does	MAIIC adoption
Indicator codes on the series	Each macro variable carries the external codes that identify it at each source (`external_codes`, for example `{"world_bank": "NY.GDP.MKTP.KD.ZG"}`), seeded once. A series with no code says so on the screen; it is not an error	Two columns added to `macro_statistics`: `external_codes` (JSON) and `country` (ISO3, default `MWI`); a seeder with the Malawi codes of 13.4
World Bank fetcher	Pulls an annual series for a country from `api.worldbank.org/v2/country/{ISO3}/indicator/{code}`: public, no key, 60-second timeout, two retries, year range optional, country from configuration with a screen override. Returns a normalised preview; writes nothing	`WorldBankFetcherService` ported as it is; `services.worldbank.country = MWI`
IMF WEO parser	Reads the World Economic Outlook download (a tab-delimited UTF-16 text file despite its `.xls` name), filters to the country and indicator, and marks each year actual or forecast from the "Estimates Start After" column. This is where the forecast periods come from	`ImfWeoParserService` ported; the file is uploaded by hand from the IMF site, which has no key-free API
Preview, then commit	Two endpoints per source: a preview the viewer may call, and a commit that writes, for the manager only. Nothing reaches the table until a person has seen the rows	Routes `macro-statistics.preview-worldbank`, `.import-worldbank`, `.preview-imf-weo`, `.import-imf-weo`; permissions `macro.view` and `macro.manage` added to the role set
Provenance per import	Every commit is a batch: source, address, fetched-at, who, how many rows, the file's hash where there is one; every observation points to its batch	`macro_source_import_batches`; `source_import_batch_id` on `macro_statistics_data`; the free-text `source` kept for manual entries
Upsert, never duplicate	Observations are keyed on series, period and period type, so a refresh updates the value and keeps the key	MAIIC's unique key on series, period and scenario already does this; an API import writes to the base scenario
A command	`macro:import-worldbank {--code=}` imports every series with a code, non-fatal per series, for the bootstrap and the scheduler	The same command; run by `eir:bootstrap` and monthly by the scheduler; a committed JSON snapshot under `docs/bootstrap/macro/` is the offline fallback so a server without internet still bootstraps
The screen	One page with five tabs: Dashboard, Variables, Data Entry, governed assumptions, Import / Export (World Bank with country and year range, IMF upload, CSV template and export)	Data Foundation, Macro Statistics (section 11.3), five tabs; the governed-assumptions tab is **Scenario Assumptions**, the base, mild and severe paths that feed Scenario Profiles and the Weighted Forecast, with the same approval lineage

Two things the suite does not have are added for MAIIC. **Reserve Bank of Malawi series by file**: the policy rate and the prime lending rate are not on the World Bank; the PLR is already in the landing zone (`P2_06`) and the policy rate arrives as a small CSV through the same preview-and-commit screen, one more source tab rather than a special case. **A rule for disagreement**: a Governance Centre setting, `macro_source_precedence`, says which source wins where two overlap.

13.4 The Malawi series, seeded

Code	Series	World Bank indicator	Why MAIIC needs it
GDP_GROWTH	Real GDP growth, annual %	NY.GDP.MKTP.KD.ZG	The headline driver of default rates
CPI	Inflation, consumer prices, annual %	FP.CPI.TOTL.ZG	Real income of borrowers; the policy-rate path
MWK_USD	Official exchange rate, MWK per USD	PA.NUS.FCRF	Import-dependent borrowers; the industrial book
LENDING_RATE	Lending interest rate, %	FR.INR.LEND	Debt service burden; cross-check to the PLR series
REAL_RATE	Real interest rate, %	FR.INR.RINR	Affordability after inflation
PRIVATE_CREDIT	Domestic credit to private sector, % of GDP	FS.AST.PRVT.GD.ZS	Credit conditions
AGRI_GROWTH	Agriculture, forestry and fishing value added, annual growth %	NV.AGR.TOTL.KD.ZG	MAIIC's book is agricultural and agro-industrial; the harvest is its own driver
BROAD_MONEY	Broad money growth, annual %	FM.LBL.BMNY.ZG	Liquidity conditions
RESERVES	Total reserves in months of imports	FI.RES.TOTL.MO	Exchange-rate and import-cover stress
DEBT_GDP	Central government debt, % of GDP	GC.DOD.TOTL.GD.ZS	Sovereign stress and crowding out
CURRENT_ACCOUNT	Current account balance, % of GDP	BN.CAB.XOKA.GD.ZS	External balance
POLICY_RATE	RBM policy rate, %	by file (RBM)	The rate the PLR follows
PLR	Prime lending rate, %	the landing zone, P2_06	The floating-rate reference of the EIR engine, so the ECL and EIR modules read one series

The IMF WEO gives the forecast years for GDP growth, inflation, the current account and government debt; the World Bank gives the actuals. A code that the World Bank later retires is a seeder change, not a code change.

13.5 Behaviour, stated precisely

- **Country.** MWI by default; the screen may fetch another country for comparison, and such rows carry their country and are never used by the FLI model.
- **Periods.** World Bank and WEO series are annual; MAIIC's FLI model runs on the frequency the series definition states. An annual value is held as the year's observation; the Weighted Forecast interpolates to quarters or months as it does today, and says so on the row.
- **Actual against forecast.** A World Bank value is an actual. A WEO value is an actual up to the estimates-start year and a forecast after it; a later WEO vintage replaces an earlier forecast and the earlier one is kept in the batch history. A forecast is never overwritten by hand without a reason.
- **Precedence** (macro_source_precedence, section 4.2): World Bank for actuals, IMF WEO for forecasts, the RBM file for rates; a manual entry overrides any of them only with a reason, and the override is shown on the row and in the audit log.
- **Failure.** A series the source cannot supply is reported on the preview and skipped on the command; the import never fails as a whole because one indicator is missing. A fetch that cannot reach the source after the retries says so and leaves the table as it was.

- **Provenance to the audit workbook.** The IFRS 9 impairment workbook of section 12 cites, for B5.5.49 to B5.5.54, the batch behind each series in use: source, address, fetched-at, who committed it.

13.6 What changes in the code

Change	Where
`external_codes`, `country` on `macro_statistics`; `macro_source_import_batches`; `source_import_batch_id` on `macro_statistics_data`	Two migrations
`WorldBankFetcherService`, `ImfWeoParserService`, `SourceImportBatch`, the RBM file parser	`app/Services/Macro`, `app/Models`
`MacroExternalCodesSeeder` with the series of 13.4	`database/seeders`
Preview and commit endpoints for the three sources; `macro.view` and `macro.manage`	`MacroStatsController`, `routes/web.php`, the permission seeder
`macro:import-worldbank`; a step in `eir:bootstrap`; a monthly schedule; the JSON snapshot	`app/Console/Commands`, `routes/console.php`, `docs/bootstrap/macro/`
The Macro Statistics screen with five tabs in the suite layout	`resources/js/Pages/FLI/MacroStats`
`macro_source_precedence` in the Governance Centre catalogue	`GovernanceService::catalogue()`
Tests: the fetcher against a recorded response, the WEO parser against a sample file, preview writes nothing, commit writes a batch, the command skips a missing series	`tests/Feature/Macro`

Not changed: Scenario Profiles, Weighted Forecast, Credit Loss Data, Adjusted Forecast and Regression Analysis read the same tables as before; the FLI model's arithmetic is untouched.

13.7 Acceptance

1. Every series of 13.4 with a World Bank code previews and imports for Malawi, and the rows carry a batch with source, address and time.
2. A WEO file previews with each year marked actual or forecast, and commits with the forecast years flagged.
3. A preview writes nothing; a commit by a viewer is refused; a commit by a manager is audit-logged.
4. `macro:import-worldbank` with the network unavailable reports every series skipped and leaves the tables unchanged; `eir:bootstrap` falls back to the snapshot.
5. The Weighted Forecast and Regression screens show the same results on imported data as on the same values typed by hand.
6. The impairment workbook's forward-looking rows cite the batches in use.

13.8 Order of work

MS-1 the migrations, the ported services, the seeder and the command, with tests (one day); MS-2 the screen in the suite layout, with the RBM file tab (one day); MS-3 the bootstrap step, the snapshot, the schedule and the audit-workbook rows (half a day). MS-1 can start with P4b; MS-2 is the first Data Foundation screen rebuilt in the suite shape and sits with UI-2.

14. Forward-looking adjustments: the regression, the correlation finder and the manual route

Since 13 October 2026 the forward-looking adjustment is specified in its own document, `MAIIC_FLI_Module_Specification_v1_2026-10-13.md` *(decisions D-F1 to D-F12 and its section 12, what is built). Sections 14 and 15 here record how the design began; where the two differ, the FLI specification governs.*

14.1 Background: why the economy enters the loss estimate

An expected credit loss is, for each loan, the probability that the borrower defaults (PD) times the share of the exposure that would be lost if they did (LGD) times the exposure at that moment (EAD), discounted at the loan's effective interest rate. Of the three, the PD is the one the economy moves most. MAIIC measures its PDs from its own history through the transition matrices: how often, over the years observed, a loan in a given grade moved to default. That history is the honest starting point, but it is an average over the years that happened to be in the window, good and bad together: a "through-the-cycle" figure. IFRS 9 asks for the probability that applies to the twelve months, or the lifetime, that is now ahead: a "point-in-time" figure that reflects where the economy is and where it is expected to go (5.5.17(c), B5.5.49). The step from the first to the second is the forward-looking adjustment.

The link between the economy and defaults is made with a **regression**: a statistical fit that says how much a credit-loss measure (the share of loans defaulting in a quarter, the NPL ratio, the roll rate from one grade to the next) has moved, historically, for a given move in an economic series (growth, inflation, the exchange rate, the harvest). Three things matter in practice. The relationship is usually **lagged**: a bad harvest shows up in defaults two or three quarters later, not in the same quarter. It is often in **changes** rather than levels: it is the fall in growth, not the level of growth, that moves defaults. And it must make **economic sense**: a fit that says defaults fall as unemployment rises is a statistical accident, however high its R-squared, and must not be used. The regression produces a predicted loss measure for each future period under an assumed path of the economy; the ratio of that prediction to the base period's prediction is the adjustment applied to every loan's PD.

Where the statistics are thin, which they are for an institution of MAIIC's size with two years of core-banking history, the standard allows judgement, provided it is reasonable, supportable and documented (B5.5.52): a management overlay. The sections that follow give both roads and the governance that makes either one auditable.

14.2 In plain language

A probability of default measured from MAIIC's own history says how often borrowers like this one have defaulted in the past. IFRS 9 asks for something more: the probability that applies to the economy that is coming, not the one that has been. The "forward-looking adjustment" is the step that turns the historical probability (the pre-FLI PD) into the one the expected credit loss is calculated on (the post-FLI PD). MAIIC's system already does this, and does it in the right order: the adjustment is worked out from a regression of credit losses on the economy, applied to each loan's PD, and the ECL reads the adjusted figure. What is missing is the discipline around it: a way to find which economic series actually explains MAIIC's losses, a test that the relationship makes economic sense before it is used, a way for Finance to apply an adjustment by judgement when the statistics cannot, and a record on each loan of where its adjustment came from. Decision D28 adds those four things and changes none of the arithmetic that already works.

14.3 How the adjustment works today

The chain, as the code runs it:

Step	What happens	Where
1	A **regression** relates a credit-loss proxy (an observed default rate, an NPL ratio) to one or more macro series over a training window; the model stores its coefficients, R-squared and window, and can be marked approved	`RegressionService`, `regression_models`
2	For a reporting period and a scenario set, a **parameter record** names the macro statistic, the PD proxy, the base period's values and a single slope and intercept	`fli_reporting_periods_parameters`

Step	What happens	Where
3	For each **forecast window** (0, 1, 2 … periods ahead) the scenario-weighted macro value is entered; the predicted proxy is slope × value + intercept; the **adjustment** is the predicted value divided by the base window's predicted value, less one: a relative uplift on the base period	`fli_adj`
4	**Applied to loans**: Stage 3 is set to 100 percent; Stage 1 takes the 12-month window; Stage 2 takes the window nearest its remaining life; post-FLI PD = pre-FLI PD × (1 + adjustment), floored at 0 and capped at 100 percent	`ExternalCalculationsController::applyToLoans`
5	The **ECL** reads the post-FLI PD; the ECL recalculation is forbidden from writing it back, because the FLI engine owns it	`TimePhasedEclService`, `RecalculateEcl --pd=pd_post_fli`

Steps 4 and 5 are sound and tested and are kept exactly as they are. The weaknesses are in steps 1 to 3:

- **One variable is applied even when several were trained.** The parameter record holds one slope and one intercept, so the model that reaches the loans is univariate whatever the approved model was.
- **No test of sense.** A model can be approved whose coefficient has the wrong sign (losses falling as unemployment rises) or whose R-squared is negligible; nothing stops it.
- **No lags, no transforms.** The economy of this quarter is regressed on the losses of this quarter; the link is usually lagged, and often in changes rather than levels.
- **Only one road.** The only way to a post-FLI PD is the regression. There is no screen on which Finance can say "for this quarter, plus fifteen percent on Stage 2 agriculture, because of the drought, approved by two people", which is what a small institution actually needs and what B5.5.52 contemplates.
- **No finder.** The analyst must already know which series to regress on.
- **No lineage on the loan.** The loan row holds the number, not which model, parameter set and scenario produced it.

14.4 The correlation finder

The finder answers the first question an analyst has: which economic series, at what lag, in what form, explains MAIIC's losses, and how well. It is the Dupleix suite's "auto-correlate", ported.

- **The sweep.** Every active macro series (section 13) against every credit-loss proxy MAIIC holds (observed default rate by product group, NPL ratio, the roll rates behind the transition matrices), across a governed lag grid (0, 3, 6, 9 and 12 months) and three transforms (level, change, change in logarithm). For each combination it computes a rank correlation first (Spearman, which is robust to the odd year) and a robust slope (Theil-Sen), then Pearson's r and a one-variable least-squares fit.
- **The ranking.** Each pair is scored on strength, sign agreement with what economics expects, overlap length and whether the series can be forecast (a series no source projects cannot drive a forward-looking adjustment, and is ranked down for it). One suggestion per pair at its best lag, with a plain-language reason: "Agriculture value added, lagged two quarters, change: Spearman −0.71 over 28 quarters; sign as expected; forecastable from the IMF".
- **Fail-closed.** A pair with too little overlap is rejected with the reason, never fabricated; a sweep with no proxy data produces no suggestions and says so.
- **Immutable runs.** Every sweep is an analysis run with the hash of the inputs it read, so a suggestion can be reproduced from the data vintage; a later sweep is a new run, not an edit.
- **The tests, governed.** Two thresholds from the Governance Centre decide what may go forward: the **expected sign** per pair (positive, negative or not stated) and an **R-squared cut-off**. A pair that fails either is shown in red with the reason, as the earlier Dupleix calculator did it; it may still be studied, but it cannot be approved as a model.

Four further pieces of the suite's forward-looking module are adopted by name, because each closes a gap a reviewer would otherwise find:

- **The guardrail that declines.** A fitted relationship may feed an adjustment only when all four hold: enough observations (a governed minimum), the realised sign matches the expected sign, R-squared at or above the governed cut-off, and statistical significance (p-value at or below a governed alpha). Otherwise the verdict is *declined* with its reason, the relationship is quarantined, and the PD stays at its pre-FLI value until a person decides. This is the sign and cut-off test of the earlier calculator made complete: a weak or wrong-signed fit cannot reach a loan by accident, and the system says so rather than silently applying nothing.
- **The structural-events register.** A governed table of the dated events that break economic series: for Malawi the 2012 float, the 2016 drought, the 2023 devaluation, the 2024 to 2025 policy-rate steps, each with its source (RBM, IMF). The finder's diagnostics test for a break at those dates (a Chow test) before trusting a fit across them; the scenario set of section 15 uses the same events as its anchors; the SICR engine may use them as qualitative triggers. One register, three consumers, so the "why 2023" of a scenario and the "why this fit fails across 2023" of a regression cite the same row.
- **The credit-loss proxy deriver.** The Y series are derived from MAIIC's own data and never typed or invented: the NPL ratio (Stage 3 exposure over gross exposure) per period from the loan books, and the Stage 1 to default 12-month rate from the approved transition matrices. With fewer than two periods a proxy is recorded as not derivable, not interpolated.
- **The series profiler.** The distribution of every X and Y series is stored, not recomputed: observations, span, mean, standard deviation, skewness and kurtosis, a normality verdict against governed limits, and a unit-root verdict, with the correlation method the shape can support. The finder reads the profile to choose rank or linear methods, and the audit workbook cites it, so the choice of method is a recorded fact rather than a default.

These add three settings to the Governance Centre beside the two of 14.8: the minimum observations for a fit (seeded 12), the significance level (seeded 5 percent), and the skewness and kurtosis limits for a normality verdict (seeded 1.0 and 3.0). The register is seeded with the Malawi events above as proposals for Dr Thom to confirm.

14.5 The regression, repaired

- **The model that is trained is the model that is applied.** The parameter record stores the approved model's id and its full coefficient vector; the predicted proxy uses every variable in it, each at its own lag and transform.
- **Approval needs the tests and two people.** A model may be approved only if it passes the sign and cut-off tests of 14.4, and approval is maker-checker: the person who trained it cannot approve it. An approved model is versioned; a retrained model is a new version and the old one stays attached to the periods that used it.
- **Back-test on the row.** Each period, the model's predicted proxy for the period just ended is compared with the realised proxy and the difference stored, so a model that has stopped working is visible before it is reused.

14.6 Three routes to the adjustment, one governed choice

A Governance Centre setting, `fli_adjustment_route`, says how the adjustment is produced for a period. All three routes write the same adjustment rows, so steps 4 and 5 of 14.3 do not change.

Route	What it is	When it is the right one
Regression (seeded default)	The chain of 14.2 with the repairs of 14.5: the approved model, the scenario paths of section 15, the predicted proxy per window, the relative uplift	When an approved model exists and passes its back-test

Route	What it is	When it is the right one
Manual overlay	A screen on which a reviewer enters the adjustment per scenario and window and, optionally, per product group or stage, with a reason and an attachment; a second person approves; the overlay has an owner and an expiry date	When no model is approvable, or when an event the statistics cannot see (a drought, a devaluation, a policy change) has to be reflected now
Regression plus overlay	The regression result, then a manual adjustment on top; each shown separately on the loan and in the ECL	When the model is sound but judgement says it is not enough

The manual overlay is a register, not a free field: every entry carries its scope, reason, evidence, owner, expiry, proposer and approver, and the ECL shows the overlay as its own line, which is how the auditor and the Board see what judgement added.

14.7 How the adjustment reaches the PD: the transmission method

The route of 14.6 says where the adjustment comes from. The transmission method says how it moves a loan's PD once it exists. MAIIC has one method today, the multiplicative scalar; it is kept as the default, and the alternatives are built as governed choices with the data each one needs, so that the method can change when the history can support it without a change to the code.

What the scalar does, and what it implies. The adjustment is a ratio to the base period (predicted proxy in the window over predicted proxy today, less one), and the loan's PD is multiplied by one plus that ratio. It is proportional, the same for every loan whatever its starting PD, and linear in the macro variable. Real credit risk is neither: a downturn moves weak borrowers far more than strong ones, and loss rates accelerate as conditions worsen. The first effect is why the cap at 100 percent is needed at all; the second is recovered at the scenario level by weighting the ECL across scenarios (15.5) even though the transmission inside each scenario stays linear. With two years of core-banking history, the scalar is the honest choice: the richer methods below need a default-rate series the institution does not yet have.

The governed setting, `fli_transmission_method`, with the preconditions the guardrail checks before a method may run. A method whose preconditions are not met is declined with the reason and the PD holds, exactly as a failed fit is declined.

Method	Transmission	Needs	Status
Multiplicative scalar on the proxy ratio (seeded)	post-FLI PD = pre-FLI PD × (1 + adjustment); the adjustment is predicted proxy(window) / predicted proxy(base) − 1	An approved fit, or an overlay	MAIIC's method; kept
Segment-specific scalar	The same, with one fit and one adjustment per portfolio or product group, from the segment proxies the deriver produces	A derivable proxy per segment (two or more periods each)	The natural next step; the deriver of 14.4 makes it possible
The nine reference methods	The suite's governed producers of a factor from a fit crossed with the scenario-weighted driver: the weighted statistic; the annual difference and the annual change in the driver; both scaled by the correlation; the absolute PD forecast from the equation; the forecast-over-base ratio (MAIIC's scalar is this one); and the two correlation-scaled difference and change variants	An approved fit with slope, intercept and correlation; the scenario-weighted driver	Ported as they are; selectable one at a time; each returns nothing rather than a fabricated figure when an input is undefined

Method	Transmission	Needs	Status
Logit-linear PD model	logit(PD) = α + β × macro; the move is in log-odds, so proportional in the odds rather than the probability and bounded by construction	A grade-level or loan-level default series long enough to fit (the guardrail's minimum observations, at the grade level)	Available when the history allows; declined until then
Vasicek single-factor Z-shift	The through-the-cycle PD is moved through the latent systematic factor: PIT PD = $\Phi[(\Phi^{-1}(\text{TTC PD}) - \sqrt{\rho} \times Z) / \sqrt{(1 - \rho)}]$, with Z from the macro fit and ρ governed; the shift is largest for mid-range PDs and bounded by construction	A governed asset correlation ρ per portfolio and a Z series calibrated on a default-rate history	Available when the history allows; declined until then

Whichever method is in force, three things do not change: Stage 3 is 100 percent; Stage 1 takes the 12-month window and Stage 2 the lifetime window; and the result is floored at 0 and capped at 100 percent. The method runs once per scenario under the weighting of 15.5, and the loan row records the method beside the route (14.8), so two loans adjusted under different methods in different periods can both be explained.

The system explains each method, in the screen, not only in this document. On the FLI Adjustments screen and beside the setting in the Governance Centre, every method has a **method card** with the same five parts: what it does, in two or three plain sentences a finance officer can read; the formula, written out with its symbols named; what it implies (proportional or not, bounded or not, linear or not, which loans it moves most); its **preconditions, checked live against MAIIC's data** at the moment the card is opened, each shown as met or not met with the figure that decides it ("default-rate series: 24 months held, 36 needed"; "asset correlation: not yet governed"); and a **worked example** on one real loan of the book, showing the pre-FLI PD, the inputs, the arithmetic and the post-FLI PD under that method. A method whose preconditions are not met cannot be selected, and the card says exactly what has to be true before it can be. The same cards are reproduced in the help centre and in the impairment audit workbook, so that the choice of method is explained in the same words to the user, the Board and the auditor. The texts are seeded content, versioned like the help articles, not strings in the code.

Switching. A change of method is a governed change with an effective date, under maker-checker, like every other setting. The first period run under a new method shows the ECL under both methods and the difference, which is itself a disclosure the auditors will want, as it is for the scenario weighting.

14.8 Lineage on the loan

Five columns are added to the loan-book row beside the adjustment and the post-FLI PD: the route used, the transmission method, the parameter record, the model version and the scenario set. The ECL report and the impairment audit workbook can then say, for every loan, which model, which overlay, which scenario and which approval produced its post-FLI PD. The scenario-weighted macro value of step 3, typed today, becomes computed and shown: the row says which scenarios, at which weights, gave the figure, and the three values behind it.

14.9 Where it lives, and the settings

Financial Modelling › Forward-Looking Model: **Correlation Finder** (new), **Regression Analysis** (repaired), **FLI Adjustments** (new: the route, the regression result, the overlay register, the apply-to-loans step with its counts), **Weighted Forecast** (now computed from the scenario set). The three settings in the Governance Centre (section 4.2): `fli_adjustment_route` (seeded: Regression), `fli_expected_sign_test` (seeded: Required), `fli_r2_cutoff` (seeded: 30 percent, a common starting threshold for annual macro data; MAIIC may tighten it). The finder's runs, the approved models and the overlays are cited in the impairment audit workbook's rows for B5.5.49 to B5.5.54.

14.10 Acceptance and order of work

1. The finder sweeps every series and proxy and ranks suggestions with reasons; a pair with fewer than the governed minimum of overlapping periods is rejected with the reason.
2. A model that fails the sign or cut-off test cannot be approved; approval needs a second person; the applied prediction uses every coefficient of the approved model.
3. Each route produces adjustment rows; under the seeded method the loans' post-FLI PDs equal pre-FLI × (1 + adjustment), floored and capped, Stage 3 at 100 percent, as today; each alternative method reproduces a hand calculation on a sample, and a method whose preconditions are missing is declined with the reason.
4. Every method has its card: description, formula, implications, live preconditions with the deciding figures, and a worked example on a real loan; a method with an unmet precondition cannot be selected and the card says what is missing; the help centre and the impairment workbook carry the same cards.
5. Every loan row names its route, method, parameter record, model version and scenario set; the ECL shows the overlay as its own line.
6. The existing ECL tests pass unchanged on the regression route and the seeded method.

FL-1 the finder with the guardrail, the register, the proxy deriver and the profiler (one and a half days); FL-2 the regression repair with its tests (one day); FL-3 the FLI Adjustments screen, the overlay register, the route setting and the transmission-method setting with the nine reference methods ported, the logit and Vasicek methods behind their preconditions, and the method cards with live precondition checks and worked examples (two days); FL-4 lineage, the computed weighting, the back-test and the workbook rows (one day). FL-1 needs the macro series of section 13; the rest can follow P4b.

15. Economic scenarios: governance and incorporation

See the note at the head of section 14: the scenario set, its weighting and the month record are now specified in the FLI module specification.

15.1 Background: what a scenario is, and why one is not enough

A scenario is a coherent story about the economy over the next few years, written down as a path for each macro series: growth, inflation, the exchange rate, interest rates, the harvest. The forward-looking adjustment of section 14 needs such a path to predict from; a single "most likely" path would be the natural choice, and it is the wrong one. Credit losses are not symmetrical: a year that is one notch worse than expected costs far more in defaults than a year one notch better saves, because borrowers who were already stretched tip into default while those who were comfortable merely become more comfortable. Calculating the loss on the single most likely path therefore understates the expected loss. IFRS 9 recognises this and asks for an unbiased, probability-weighted estimate over a range of possible outcomes (5.5.17(a), B5.5.42): the loss under each of several scenarios, each weighted by how likely it is, and the weighted average reported.

Scenarios for the ECL are not the same thing as stress tests. A stress test asks what happens under a severe but plausible shock, with no weight attached, to see whether capital would hold; its scenarios are deliberately harsh. ECL scenarios are a probability-weighted view of what is actually expected, in which the severe case carries a small weight and the base case a large one. The two should share their economics (the same base path, the same understanding of what a devaluation does to the book) but they answer different questions, and a system that uses one set for both will be wrong for one of them.

The weights are the part that attracts the most scrutiny, because a small change in a weight can move the provision materially and nothing in the data pins them down. Good practice therefore does three things: anchors each scenario's severity to something that has actually happened, so that "a downside" is not an abstraction but "2023 again"; sets the weights by a documented judgement that two accountable people approve; and tests afterwards whether the realised year fell inside the range the scenarios spanned, so that

weights that were too optimistic are seen and revised. For Malawi the anchors are not hard to find: the 2023 devaluation of 44 percent, the 2016 drought, the 2012 float, and the 2021 to 2022 recovery each give a path that the institution's own loan book has already been through.

15.2 In plain language

The forward-looking adjustment depends on what the economy is assumed to do. IFRS 9 does not let that be one guess: it asks for an unbiased, probability-weighted view across a range of possible outcomes (5.5.17(a), B5.5.42). In practice that means a base view, a better one and a worse one, each with a story, a set of figures and a weight, agreed by the people accountable for it and kept once the period's numbers are struck. MAIIC's system holds scenarios today, but in two unconnected places, without a period, a version, a narrative, a source, an approval or a lock, and it uses them at the wrong point: it averages the economy and calculates one loss, where the standard asks for the loss under each economy, then the average. Decision D29 makes the scenario set a governed object and moves the weighting to where it belongs.

15.3 What good practice asks for

Principle	Where it comes from	What it means for MAIIC
Unbiased and probability-weighted, over a range of outcomes	IFRS 9 5.5.17(a), B5.5.42	At least a base, an upside and a downside, with weights that sum to 100; the ECL is the weighted average of the ECL under each scenario, not the ECL at the average economy
Reasonable and supportable, without undue cost or effort	B5.5.49 to B5.5.54	Paths from published sources (the IMF World Economic Outlook, the World Bank, the Reserve Bank of Malawi), with the vintage recorded; judgement written down where the sources stop
Governed	The Basel Committee's guidance on credit risk and accounting for expected credit losses, principles 2 and 5; the auditors' expectation	Approved by two people, one from Finance and one from Risk; locked with the period's ECL; changed only by a new version with a reason; reviewed at least once a year
Severity calibrated to history	Common practice	The downside is anchored to what Malawi has lived through: the 2023 devaluation, the 2016 drought, the 2012 float; the set says which history each scenario is calibrated to
One house view	Basel guidance; audit consistency	The same scenario set drives the ECL, the sensitivity disclosures, the stress test and the budget; a different set for each is a finding
Disclosed	IFRS 7.35G	The narratives, the weights, the key paths per scenario, the ECL under each scenario and with 100 percent weight on each, and the sensitivity to the weights
Overlays governed, not hidden	Basel guidance; B5.5.52	A management overlay sits on an approved set, with a reason, an owner, an expiry and a second approval, and is shown separately
Back-tested	Model-risk practice	Each period the previous set's base path is compared with what happened, and the predicted default rate with the realised; a miss outside the range triggers a review of the weights

15.4 The scenario set, as a governed object

The two existing structures (`scenario_profiles` with its child scenarios, and `scenario_sets` with `scenario_probabilities`) become one, and a migration maps what is there.

The set, one per reporting period: period; name; version; status (draft, proposed, approved, locked); the narrative; the source vintage ("IMF WEO April 2026; World Bank 2025 actuals; RBM Monetary Policy Committee, March 2026"); proposed by; approved by (two people); locked at. A set is locked when the period's ECL is locked; a change after that is a new version with a reason, and the locked version stays attached to the period.

The scenarios under it, three or more: name; probability weight (the weights must sum to 100); the narrative; the calibration note that says which history the scenario is anchored to; and the **paths**: for each macro series and each horizon, the base scenario's path comes from the sources of section 13, and every other scenario is expressed as a governed **shock** on the base, by series and year offset, of one of four kinds: a percentage change, an absolute change, a replacement value, or a multiplier. A downside is therefore an auditable transformation of the base, not a second set of typed numbers; change the base and the shocked paths follow.

15.5 Weighting the loss, not the economy

A Governance Centre setting, `scenario_weighting_method`, with two options:

Option	Meaning
Weight the ECL across scenarios (seeded)	The forward-looking chain of section 14 runs once per scenario; each loan carries a PD and an ECL under each scenario; the reported ECL is the probability-weighted sum. This captures non-linearity: a downside hurts more than an upside helps, and the average of the losses is higher than the loss at the average
Weight the macro path, one ECL	Today's method: the scenario-weighted macro value, one regression, one ECL. Kept so that past periods can be reproduced and the two methods reconciled when the switch is made

Switching is a governed change with an effective date; the first period run under the new method shows both figures and the difference, which is itself a disclosure the auditors will want.

15.6 The rules, as settings

In the Governance Centre: the minimum number of scenarios (seeded 3); a floor on the base scenario's weight (seeded 40 percent) and a ceiling on any single weight (seeded 60 percent); whether a calibration note is required on every downside (seeded Required); the annual review month; and the rule that an overlay needs an approved set (seeded Required). Each is changed under maker-checker with a reason, like every other setting.

15.7 Overlays, back-tests and sensitivity

- **The overlay register** of section 14.5 is tied to the scenario set: an overlay names the set it sits on, and a set cannot be locked with an unapproved overlay against it. Each overlay is shown as its own line in the ECL and in the disclosure.
- **Back-test, every period**: for each macro series, the previous set's base path against the actual now known; for each proxy, the predicted default rate against the realised. The results are stored with the set, and a miss outside the scenario range raises a flag that the weights are to be reviewed before the next set is proposed.
- **Sensitivity, every period**: the ECL under each scenario; the ECL with 100 percent weight on each; the ECL with ten points moved from the base to the downside and to the upside. Stored with the set, so the IFRS 7.35G tables are a report from the system, not a spreadsheet beside it.

15.8 The first set, proposed

For MAIIC the first set is seeded as a proposal for Dr Thom, every scenario anchored to a year Malawi has lived through:

Scenario	Weight	Anchored to
Base	50	The IMF World Economic Outlook path, World Bank actuals, the RBM's stated policy path
Upside	15	The 2021 to 2022 recovery: growth and the harvest above the base, inflation below
Downside	25	The 2023 devaluation year: the kwacha down by 44 percent, inflation and lending rates up
Severe	10	The 2016 drought together with a devaluation: the harvest fails and the currency falls

The weights are a starting point and are his to set; the anchors are the answer to the question "why these".

15.9 Where it lives

Governance Centre › **Scenario Sets** (create, propose, approve, lock, versions, history, the back-test and sensitivity stored with each); Data Foundation › Macro Statistics › **Scenario Assumptions** (the base paths and the shocks); Financial Modelling › Forward-Looking Model › **FLI Adjustments** (the chain run per scenario); Report Hub › **IFRS 9 Disclosure** (the 7.35G tables) and the impairment audit workbook, which carries one row per principle of 15.3 citing the set in force.

15.10 Acceptance and order of work

1. A set cannot be proposed with weights that do not sum to 100, fewer scenarios than the minimum, or a downside without its calibration note; it cannot be approved by its proposer; it cannot be changed after lock except as a new version with a reason.
2. Every non-base path equals the base path transformed by its recorded shocks, and changing the base re-derives them.
3. Under the seeded weighting method, every loan carries a PD and an ECL per scenario and the reported ECL is the weighted sum; under the other method the figure equals today's.
4. The back-test and the sensitivity tables are produced for a locked period and agree with a hand calculation on a sample.
5. The disclosure report prints the narratives, weights, paths, per-scenario ECL and weight sensitivity of the set in force.

SC-1 the set, the scenarios, the shocks, the migration from the two old structures and the governance rules (two days); SC-2 the ECL per scenario and the weighting method (one day); SC-3 the overlay tie, the back-test, the sensitivity and the disclosure tables (one day); SC-4 the first set seeded as a proposal (half a day). SC-1 follows section 13; SC-2 follows FL-3.

16. The Mega Farm loans, seed loans included: a full treatment

16.1 Background: what the Mega Farm programme is

In 2024 the Government of Malawi set aside K20 billion to lend to farmers for the things a season needs: fertilizer, seed, pesticides, working capital, equipment and irrigation. It asked MAIIC to run the programme. MAIIC finds the farmers, approves the loans, pays them out (in 2025 mostly as vouchers for farm inputs rather than cash), keeps the accounts and collects the repayments. Many farmers repay not in money but in maize, delivered to one of three buyers (ADMARC, NFRA or ACE), who then owe MAIIC the value of the maize.

The loans carry interest at 15 percent a year. By agreement, 5 percent of that belongs to MAIIC and 10 percent belongs to the Government's fund. MAIIC also earns a management fee of 5 percent on the money the fund receives, and a commission. When a farmer does not repay, the loss is charged to the Government's fund, not to MAIIC. The 2025 financial statements show the fund absorbing K37.4 billion of expected losses and a K1.2 billion write-down on maize, and closing the year at K16.0 billion.

In E-Banker the programme lives in eight schemes (96 fertilizer, 97 irrigation, 98 seed, 99 CAPEX, 100 working capital, 101 pesticides, 102 and 103 equipment) with about 7,000 accounts. They have their own

general-ledger codes: the loans on 1050301 to 1050307, the interest owed by farmers on 1059 to 1067, the income on 4214 to 4224 and 4260 to 4262, the fund on 2070 to 2072. None of these schemes was in the 18 that the 7 October extracts covered, so we hold no postings, balances or loan-book runs for any Mega Farm account yet; pack 2 of 8 October fetches them.

16.2 What the December 2025 financial statements say

The statements keep the programme separate from MAIIC's own lending on every page. On the balance sheet the loans, the cash, the maize and the fund each have their own "Mega Farms" line. Note 8 gives them their own table, which uses the three "stages" of IFRS 9. In plain words: Stage 1 is a loan that is performing, Stage 2 is a loan whose risk has gone up significantly since it was made, and Stage 3 is a loan that is in default. The "expected credit loss" is the provision, the amount set aside because some of the loan will not come back.

Mega Farms, 31 December 2025 (K thousand)	Stage 1	Stage 2	Stage 3	Total
Gross loans	2,799,359	nil	48,744,898	51,544,257
Expected credit losses	(1,162)	nil	(39,760,244)	(39,761,406)
Net	2,798,197	nil	8,984,654	11,782,851

So 95 percent of the programme is in default, and 82 percent of the defaulted amount has been provided for. The seed loans are K7.59 billion of the K51.54 billion (code 1055 in the accounting system, 1050302 in E-Banker), with their own interest-owed account (1062, K0.79 billion after the audit adjustments) and their own income line (4217, K412 million).

Three adjustments made during the audit show how the interest is handled:

- **"Transfer of MAIIC interest to receivable."** MAIIC's 5 percent share of the interest was moved out of the Mega Farm interest accounts into MAIIC's own "other receivables". Note 9a describes it: "interest income from Mega farms amounting to K2.8 billion, which carries expected credit losses of K2.1 billion". For seed, that transfer was the whole of the K412 million.
- **The fund's 10 percent share** is credited to the fund (code 2071, K6.4 billion).
- **"Fair value adjustment on maize."** The amounts owed by the three maize buyers were written down to what the maize is actually worth.

Note 19 then charges "impairments on MAIIC Mega farm interest income" of K2.1 billion to MAIIC's own profit. That is the only Mega Farm loss MAIIC itself bears; the rest is the fund's.

The monthly trial balances tell the same story from the other side. The seed loan balance did not change at all from January to June 2025 and was repaid only in the harvest months. Interest is added to a separate "owed" account, never to the loan itself. The income on seed works out at 4.6 percent of the balance, which is MAIIC's 5 percent share, not the 15 percent the farmer pays. And K350 million left the seed interest-owed account in June 2026 without passing through income, which is the same "transfer to receivable" being done during the year.

16.3 Whose loan is it, explained

Before asking what interest rate a loan earns, an accountant asks whose loan it is. The accounting rules put the question in two ways. First, does the lender carry the risks and rewards of the loan: does it lose if the borrower fails, and gain if the borrower pays? Second, is the lender acting for itself (a "principal") or on behalf of someone else (an "agent")? The statements say Deloitte looked at exactly this question for the programme (the note on IFRS 10 and "de facto agents", page 34).

For the Mega Farm loans the answers point the same way:

- **The risk is the Government's.** When a farmer fails to pay, the loss comes off the Government's fund. MAIIC loses only its own 5 percent interest share, which is why that share is moved to MAIIC's own receivables and provided for there.
- **MAIIC's return is a fixed slice, not a yield.** The farmer pays 15 percent. MAIIC's part is 5 percent, set by the agreement, plus the fee and the commission. There are no arrangement or legal fees charged to the farmer that an effective interest rate would have to spread, and the rate does not move with the prime rate.
- **MAIIC does the work and keeps the accounts.** It approves, pays out, collects and stages the loans, and the statements show the loans on MAIIC's balance sheet with the fund as a matching liability on the other side.

The statements settle it by showing the loans on MAIIC's balance sheet but kept apart, with the fund as the other side and MAIIC's own exposure limited to its interest share. The engine has to follow that, because the auditors signed it, and it must not pour a K51.5 billion programme book into a calculation built for MAIIC's own K14.7 billion of lending.

16.4 The treatment, decision D30

Question	Treatment	Why, in plain words
Are the Mega Farm loans in the EIR engine?	**No.** They stay outside, and every engine report and audit workbook says so	The effective interest rate exists to spread a loan's fees and discounts over its life. These loans have no such fees, their rate is fixed by agreement and split by contract, and MAIIC's return is 5 percent whatever happens. Recalculating would change nothing the statements report
Is MAIIC's 5 percent share computed?	**Yes.** The engine works out 15 percent on what is actually expected to come back from each loan (the balance after provisions), splits it 5 to MAIIC's receivable and 10 to the fund	That share is MAIIC's own income; it was written down by K2.1 billion; the auditors look at it. The rules say interest on a defaulted loan is counted on the amount expected to be recovered, not on the full balance
Are the loans in the ECL module?	**Yes**, completely: staging, the chance of default, the loss if default happens, and the provision for every scheme; the provision charged to the fund, and the write-down of MAIIC's interest share charged to MAIIC	The staging and the K39.76 billion provision are MAIIC's own calculation and were audited; the module exists and already does this for the other book
Repayment in maize	The loan is treated as repaid by the value credited; what the maize buyer owes is recorded as a receivable and written down to what the maize is worth, the write-down charged to the fund	This is what note 9c does; the engine records it rather than Finance typing it
The fund	Shown as a liability (codes 2070 to 2072) and rolled forward every month: money received, the fund's 10 percent interest, new loans, repayments, write-downs, provisions	Note 11(c) is that roll-forward; the system produces it instead of a spreadsheet
How the loans are described in the system	Their own scheme settings, read from E-Banker: the 15 percent and its split, interest kept in a separate account, repayment once a year after harvest, a maize buyer as the payer	The MAIIC book's rules (interest added to the balance, moratoria, instalments) do not describe these loans

16.5 How a seed loan flows through the system, step by step

1 **Landing.** The pack 2 extracts for scheme 98 arrive through the same door as everything else: the account and loan masters, every posting, the month-end balances, the loan-book runs, the status history, the rates, the charges. The same checks apply.

2. **Build.** The seed accounts are built into the loan book by whichever method is in force, with two rules of their own: the amount owed is the loan balance plus MAIIC's share of the interest owed, and the fund's share of the interest owed is kept in its own column.
3. **Staging and provision.** Each seed account is staged and provided for by the ECL module like any other loan, using the Mega Farm scheme's own settings (its own default history, its own recovery assumptions, which allow for repayment in maize). The provision is charged to the fund.
4. **Interest.** The engine works out 15 percent for the month on the amount expected to be recovered, puts 5 to MAIIC's receivable and 10 to the fund, and compares MAIIC's share with what E-Banker actually posted. The EIR calculation proper (schedules, the solved rate, the revenue shift) is not run on these loans.
5. **MAIIC's own write-down.** MAIIC's receivable for its 5 percent is itself provided for in MAIIC's own books, as note 9a shows.
6. **Reporting.** The Mega Farms table of note 8, the fund roll-forward of note 11(c), the receivable and its provision of note 9a, and the maize of note 9c come out of the Report Hub as one Mega Farms report, so the figures the auditors sign are produced by the system rather than typed.
7. **Audit.** The IFRS 9 impairment workbook carries the Mega Farm rows; the Contract Schedule 1 workbook records that the programme is outside the EIR engine and why.

16.6 Two checks the pack 2 data will settle

The seed interest-owed account rose by K700 million in 2025 while seed income on the trial balance was K379 million. The K320 million gap should be the fund's 10 percent share, credited to the account and then passed to the fund; the postings will show it line by line. And the K350 million that left the account in June 2026 should be MAIIC's share being transferred to its own receivables during the year; if it turns out to be a write-off, that is a different conversation with Finance. Both become checks the system runs once the extracts arrive, and both go on the Baselines sheet.

16.7 What this does to the headline number

Dr Thom asked for one number: how much revenue moves between years when MAIIC's interest is recalculated at the effective rate. That number is about MAIIC's own lending: K1.4 billion of loan interest in 2024 and K5.6 billion in 2025. The Mega Farm interest, half of all loan interest in 2025 on the trial balance, is a programme return of which two-thirds belongs to the Government and the rest is largely owed but not collected; it does not change with the effective rate and stays where the statements put it. Leaving the programme out of the EIR makes the number smaller, cleaner and easier to defend, which is what it needs to be.

16.8 Probabilities of default for a book with little history

The difficulty. MAIIC measures its probabilities of default with transition matrices: how often, over the years observed, a loan in one grade moved to another, and in particular to default. That needs years. The Mega Farm programme has two: the 2024 loans (K10.6 billion, a third in default by December 2024) and the 2025 loans (K44.4 billion of new lending, almost all in default by December 2025). What it does have is breadth: about 7,000 accounts and up to 21 monthly loan-book runs each, roughly 150,000 account-months. The problem is not too few observations; it is that every observation comes from one or two seasons, one of them catastrophic. A matrix fitted to that would say the probability of default is 95 percent for ever, which describes 2025 and forecasts nothing.

How the 2025 provision was actually arrived at is still to be confirmed. The statements show the result (note 8: K39.76 billion against K51.54 billion gross) but not the method, and that method is the starting point for anything the system does. The question is with Finance (section 5); until it is answered, the treatment below is the proposal.

The method, in six steps.

1. **Measure by season, not by month.** A Mega Farm loan falls due once, after harvest, so "days past due" means nothing until the due date. The states are set at the due date plus a grace period (paid in full, partly paid, unpaid) and again at the next season's due date. A monthly matrix on a loan like this sees twelve months of "current" and then one jump, and learns nothing in between.
2. **Segment by what drives repayment.** Seed against fertilizer against equipment; voucher against cash; which maize buyer (ADMARC, NFRA, ACE, none); district; the cohort year. With 7,000 accounts a segment's default rate is well measured even from one season, and the differences between segments are real information that a single overall rate hides.
3. **Anchor the probability on cohort default rates, with a benchmark.** Each segment's observed season default rate is blended with a benchmark for a normal year (Malawi's farm-input programmes and their recovery record, microfinance agricultural lending, the 2024 cohort as the one ordinary season on record), weighted by how many seasons of own data exist. This is the standard treatment of a short-history book: the own experience gains weight each season, and the benchmark says what a normal year looks like until the book has seen one.
4. **Put the probability second and recovery first.** For the 95 percent already in default the probability is 100 percent by definition; the provision depends entirely on how much comes back: the maize delivered, its value, how much the buyers pay and when. The loss given default is built from the 2025 collection experience (K14.8 billion repaid on the fund, the maize write-down, the buyer receivables) by segment and season. That is where the K39.8 billion of provision comes from, and it is the part the data supports.
5. **Forward-looking by scenario, not by regression.** A regression on the economy needs years; two seasons give nothing. The scenario set of section 15 (a normal rainfall season, a drought season, a devaluation year) carries a default rate and a recovery rate per segment for the programme, weighted as section 15 describes, and the manual overlay route of 14.6 is used with the reason written down.
6. **Keep it separate and label it.** Under the Reserve Bank's directive (section 3.6) the programme's seasonal facilities are short term and non-performing from 91 days past due, not the 181 that applies to MAIIC's medium- and long-term book; the governed thresholds carry a Mega Farm class for that. The MAIIC book keeps its transition-matrix probabilities. The Mega Farm schemes get their own method card in the system (section 14.7's cards), their own back-test every season, and a rule that the method is reviewed once three seasons exist. The audit workbook then says exactly what was done for which book and why.

A third option: MAIIC's agricultural-sector PD, scaled to the programme. MAIIC's own agricultural book has what the programme lacks, which is time: six years of migrations, a term structure (how the probability builds over 12, 24 and 36 months) and an observed sensitivity to the same drivers, rainfall, maize prices, the kwacha and input costs. What it does not have is the programme's level of default: a sector probability of a few percent applied to a book that lost a third of its 2024 cohort in a year, and 95 percent of its 2025 cohort, would be wrong exactly where it matters, on the Stage 1 loans and on every new season. The method therefore takes the shape from the sector and the level from the programme:

probability (Mega Farm segment, horizon) = probability (MAIIC agricultural sector, horizon) × programme scalar

- The sector probability comes from MAIIC's approved agricultural transition matrix, or from the external-rating mapping for the same sector, and brings the term structure and the macro sensitivity with it.
- The scalar is the ratio of what the programme actually did to what the sector did in the same year: the 2024 cohort's 33 percent against the sector's 12-month probability, the 2025 cohort's 95 percent likewise. It is calibrated per segment where the data supports it (seed against fertilizer, voucher against cash, by maize buyer), governed, re-estimated every season and back-tested. It is a measured ratio, not a judgement.

- The forward-looking adjustment of section 14 and the scenarios of section 15 act through the sector probability, so a drought scenario moves the programme's probability the way it moves the sector's, only more.
- Guardrail: the method is declined until the scalar can be calibrated on at least one completed season, and a scalar above a governed ceiling is flagged for a second approval, so that it cannot quietly be set to one.

This is the benchmark prior of step 3 with the prior chosen well: MAIIC's own sector, on the method the auditors already accept, rather than an outside statistic. The plain cohort rate is kept beside it as the check.

The governed setting, `megafarm_pd_method`, with four options: *MAIIC agricultural-sector PD scaled to the programme* (seeded); *seasonal cohort default rate with a benchmark prior*, the check against the first; *transition matrix on the programme's own history*, available once three seasons exist and declined until then; *expert judgement with overlay*, for a season the data cannot describe. The programme scalar per segment, its ceiling, the benchmark rate and the credibility weight are governed amounts beside it, and each option has its method card (14.7) with the preconditions checked live.

What the two seasons already say. The 2024 cohort, the only ordinary season on record, lost a third in its first year. The 2025 cohort was four times larger, mostly vouchers, and almost wholly defaulted. Those two points already bracket the normal and the bad scenario of step 5, and the cohort split is the first segment to run once the pack 2 extracts arrive.

17. Glossary of the new terms

- **Narration**: the text E-Banker writes on each posting. On an interest posting it states the period and the rate, which is how section 3.1 was proven.
- **Stored loan book**: the table behind the printed Loan Book Report, one row per account per run.
- **Latest run**: of several stored runs of the same month, the one with the highest row id; the one the engine keeps.
- **Take-on**: the loading of the pre-existing loans into E-Banker on 31 July 2024, with one opening posting each.
- **The DFI directive**: the Reserve Bank of Malawi's Financial Services (Credit Risk Management for Development Finance Institutions) Directive, 2018, which classifies MAIIC's facilities by days overdue and tenor; the source of Stage 3 at 181 days (section 3.6).
- **Days past due basis**: whether days overdue are counted from the oldest unpaid instalment (the directive's count, governed default) or read from the ageing of overdue amounts by bucket.
- **Non-performing**: substandard, doubtful or loss under the directive; placed on non-accrual.
- **Diff Int by ROI**: E-Banker's year-end re-rating of a year's interest to the rate on the account at year end: debits to customers as type 303, credits as type 120, and one net voucher per batch to interest income.
- **Bootstrap (method A)**: building a month's loan book from the stored Loan Book Report's latest run, as E-Banker printed it.
- **Derivation (method B)**: building it from the ledger and the masters, with E-Banker's own arrears fields.
- **Report importer (method C)**: building it from the printed Loan Book Report uploaded as Excel, the way MAIIC has loaded every month until now, with the file landed and date-checked.
- **As at a date**: the computation of a loan's EIR and amortised cost using only what was posted, in force and locked on that date; any date, not only a month-end.
- **Take-on basis**: whether a take-on loan's EIR is recomputed from its origination (schedule and fees) or started at its 31 July 2024 balance; recorded on every take-on loan.

- **Bootstrap (the command)**: `eir:bootstrap`, which builds a clean install from the inputs committed under `docs/bootstrap/`, runs the engine chain and verifies the result against the baselines and the golden numbers.
- **Golden number**: a figure MAIIC has already accepted (an audited allowance, a tie proven on the ledger, an approved run's result) that every later build must reproduce.
- **Batch (macro import)**: the record of one import of macro statistics: source, address, fetched-at, who committed it, rows; every observation points to its batch.
- **WEO**: the IMF World Economic Outlook database, the source of the forecast years.
- **Pre-FLI and post-FLI PD**: the probability of default measured from history, and the same probability after the forward-looking adjustment; the ECL is calculated on the second.
- **Correlation finder**: the sweep of every macro series against every credit-loss proxy, over lags and transforms, that ranks which relationships are worth a model.
- **Guardrail**: the four tests (observations, sign, strength, significance) a fitted relationship must pass before it may adjust a PD; a fit that fails is declined with its reason and quarantined.
- **Structural event**: a dated break in the economy (a float, a drought, a devaluation) kept in a governed register and used by the diagnostics, the scenarios and the SICR triggers alike.
- **Transmission method**: how an adjustment moves a loan's PD once it exists: a multiplicative scalar, a segment scalar, one of the nine reference producers, a logit-linear model or a Vasicek shift; governed, and declined where its data does not exist.
- **Overlay**: a forward-looking adjustment applied by judgement rather than by a model, with its reason, owner, expiry and two approvals, shown as its own line.
- **Scenario set**: the governed collection of economic scenarios for a reporting period, with their weights, paths, narratives, source vintage and approvals.
- **Shock**: the recorded transformation that turns the base path into another scenario's path: a percentage change, an absolute change, a replacement or a multiplier, by series and year.
- **Mega Farm programme**: the Government's K20 billion farm-input lending scheme that MAIIC administers; loans at 15 percent split 5 to MAIIC and 10 to the fund; losses charged to the fund; presented ring-fenced in the financial statements.
- **Cohort default rate**: the share of the loans made in one season that had defaulted by a set point after their due date; the probability measure for a book too young for a transition matrix.
- **Programme scalar**: the measured ratio of the Mega Farm programme's default rate to MAIIC's agricultural-sector probability for the same year and horizon; the level the sector's shape is scaled to.
- **Benchmark prior**: the default rate assumed for a normal year from outside evidence, blended with the book's own experience until the book has enough seasons of its own.
- **Fund share**: the 10 percent of Mega Farm interest that belongs to the Government fund; MAIIC's share is the other 5 percent.
- **Landing zone**: the raw tables that mirror E-Banker, loaded exactly as received and never edited; everything else is derived from them.
- **Pack**: one month's set of extract files plus a manifest of what they are, which query version made them and their hashes; the one form in which data enters, whichever route delivers it.
- **Watermark**: the last source key loaded for a table; the next pack starts after it.
- **Seeded default**: the value Dupleix wrote into the Governance Centre as the first approved value, in force until MAIIC changes it.
- **Icon rail**: the navigation panel folded to a narrow column of icons; labels appear as tooltips.
- **Appearance (light, dark, follow the device)**: the per-user choice of how the screens are coloured; kept in the browser for the first paint and on the user's record so that it follows them.

- **Breadcrumb**: the line at the top of a page that reads group, then page, so the reader knows where in the system they are.
- **Audit workbook**: one Excel file per standard or directive, one row per section, with what the system does, where to see it, a status and a sign-off; built from a data module together with its Markdown and PDF twins.
- **Baselines sheet**: the sheet of the audit workbook that re-checks the acceptance ties of section 9 against the live database with a PASS or FAIL formula.
- **Auditor's pack**: the period's export, the five workbooks and a manifest of checksums, bundled for Deloitte.
- **Control exception**: a difference between the signed offer letter and the system record, reported to Credit for action; not an accounting entry.